



Neet Test Hub Premium Test 13 Solution

Physics

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11 - B	12 - A	13 - D	14 - C	15 - D	16 - B	17 - D	18 - C	19 - D	20 - C
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Chemistry

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Biology - (Zoology)

91 - A	92 - A	93 - B	94 - A	95 - C	96 - D	97 - D	98 - A	99 - C	100 - C
101 - A	102 - C	103 - A	104 - D	105 - A	106 - D	107 - C	108 - A	109 - C	110 - D
111 - C	112 - C	113 - A	114 - A	115 - B	116 - A	117 - C	118 - A	119 - D	120 - A
121 - A	122 - B	123 - A	124 - B	125 - D	126 - B	127 - B	128 - D	129 - D	130 - A
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Biology - (Botany)

136 - C	137 - C	138 - C	139 - A	140 - D	141 - C	142 - C	143 - B	144 - C	145 - C
146 - B	147 - D	148 - C	149 - B	150 - B	151 - D	152 - A	153 - C	154 - A	155 - B
156 - A	157 - B	158 - B	159 - A	160 - D	161 - B	162 - D	163 - D	164 - B	165 - B
166 - B	167 - B	168 - C	169 - A	170 - D	171 - A	172 - D	173 - D	174 - B	175 - A
176 - B	177 - C	178 - B	179 - D	180 - C					

Physics

1. A double-slit experiment is immersed in a liquid of refractive index 1.33. The separation between the slits is 1 mm and the distance between the slit and screen is 1.33 m. If slits are illuminated by a parallel beam of light whose wavelength is $6300 \text{ \AA}$, then fringe width will be

- A) 6.3 mm B) 63 mm
C) 0.63 mm D) None of these

Solution : (Correct Answer: C)

Given, $\lambda = 6300 \text{ \AA} = 6300 \times 10^{-10} \text{ m}$

$$= 6.3 \times 10^{-7} \text{ m}$$

When equipment is immersed in a liquid of refractive index 1.33, then wavelength (λ) is changed.

$$\therefore \lambda' = \frac{\lambda}{\mu} = \frac{6.3 \times 10^{-7}}{1.33}$$

$$\therefore \text{Fringe width, } \beta = \frac{D\lambda'}{d}$$

$$= \frac{1.33 \times 6.3 \times 10^{-7}}{1 \times 10^{-3} \times 1.33}$$

$$= 6.3 \times 10^{-4}$$

$$= 0.63 \times 10^{-3} \text{ m}$$

$$= 0.63 \text{ mm}$$

2. If the fringe width obtained in Young's double-slit experiment is 1.33 mm, then on immersing the whole set up in the water with refractive index 1.33, new fringe width will be

- A) Does not change B) 1.0 mm
C) 2.0 mm D) 2.69 mm

Solution : (Correct Answer: B)

$$\beta' = \frac{\beta}{\mu} = \frac{1.33}{1.33}$$

$$= 1 \text{ mm}$$

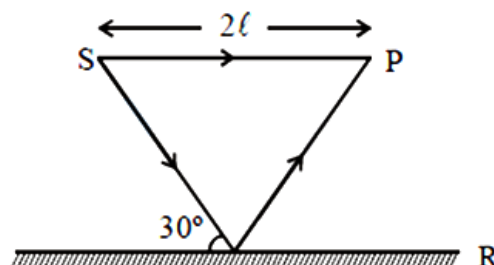
3. In Young's double-slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$, without disturbing the geometrical arrangement, the new fringe width will be

- A) 0.30 mm B) 0.40 mm
C) 0.53 mm D) 450 microns

Solution : (Correct Answer: A)

$$\beta' = \frac{\beta}{\mu} = \frac{0.4}{(4/3)} = 0.3 \text{ mm}$$

4. The interference pattern is observed at P due to superimposition of two rays coming out from a source S as shown in the figure. The value of l for which maxima is obtained at P is (R is a perfect reflecting surface): -



- A) $l = \frac{(2n-1)\lambda}{2(\sqrt{3}-1)}$ B) $l = \frac{(2n-1)\lambda}{\sqrt{3}-1}$
C) $l = \frac{(2n-1)\lambda\sqrt{3}}{4(2-\sqrt{3})}$ D) $l = \frac{2n\lambda}{\sqrt{3}-1}$

Solution : (Correct Answer: C)

$$x \cos 30^\circ = l$$

$$x \frac{\sqrt{3}}{2} = l$$

$$x = \frac{2}{\sqrt{3}} l$$

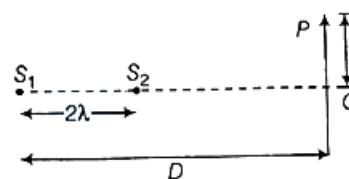
$$\frac{2l}{\sqrt{3}} + \frac{2l}{\sqrt{3}} + \frac{\lambda}{2} - 2l = n\lambda$$

$$2l \left(\frac{2}{\sqrt{3}} - 1 \right) = (2n-1) \frac{\lambda}{2}$$

$$2l (2 - \sqrt{3}) = \frac{(2n-1)\lambda\sqrt{3}}{2}$$

$$l = \frac{(2n-1)\lambda\sqrt{3}}{4(2-\sqrt{3})}$$

5. Two coherent point sources S_1 and S_2 vibrating in phase emit light of wavelength λ . The separation between them is 2λ . The light is collected on a screen placed at a distance $D \gg \lambda$ from the slit S_1 as shown. The minimum distance, so that intensity at P is equal to the intensity at O

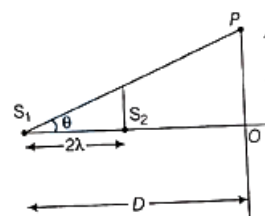


- A) $\sqrt{2}D$ B) $\sqrt{3}D$ C) $\sqrt{8}D$ D) $\sqrt{5}D$

Solution : (Correct Answer: B)

Path difference between the light waves from S_1 and S_2 , reaching at P is,

$$= 2\lambda \cos \theta$$



Since, there is a bright maximum at O. Thus for

maximum at P ,

$$2\lambda \cos \theta = n\lambda$$

For x to be minimum $n = 1$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$\therefore x = D \tan 60^\circ = \sqrt{3}D$$

6. In Young's experiment, the wavelength of red light is 7.8×10^{-8} cm and that of blue light is 5.2×10^{-8} cm. Value of n for which $(n+1)^{\text{th}}$ blue bright fringe coincides with n^{th} red fringe is

A) 1 B) 2 C) 3 D) 4

Solution : (Correct Answer: B)

As we know that $n_1 \lambda_1 = n_2 \lambda_2$

$$\therefore \frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} = \frac{7.8 \times 10^{-8}}{5.2 \times 10^{-8}} = \frac{3}{2}$$

Also $n_2 - n_1 = 1$

$$\therefore n_1 = 2$$

7. In Young's double-slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$, without disturbing the geometrical arrangement, the new fringe width will be

A) 0.30 mm B) 0.40 mm
C) 0.53 mm D) 450 microns

Solution : (Correct Answer: A)

$$\beta' = \frac{\beta}{\mu} = \frac{0.4}{(4/3)} = 0.3 \text{ mm}$$

8. A parallel beam of monochromatic light falls normally on a single narrow slit. How does the angular width of the principal maximum in the resulting diffraction pattern varies with the width of the slit?

A) Principal maximum increases on increasing the width of slit
B) Principal maximum decreases on decreasing the width of slit
C) Principal maximum increases on decreasing the width of slit
D) None of these

Solution : (Correct Answer: C)

Angular width of central maxima = $\frac{2\lambda}{b}$

$\therefore$ The angular width of the principal maximum increases on decreasing the width of slit (b).

9. Unpolarized light passes through two polaroids, the axis of one is vertical and that of the other is 30° to the vertical. What is the orientation and intensity of the transmitted light?

A) Plane polarized at 60° to the vertical and having intensity of $\frac{I_0}{4}$
B) Plane polarized at 30° to the vertical and having intensity of $\frac{3I_0}{8}$
C) Plane polarized at 30° to the vertical and having intensity of $\frac{I_0}{2}$
D) No light passes

Solution : (Correct Answer: B)

The first polaroid aligns half the light so that the intensity is reduced by half $I' = \frac{I_0}{2}$. The light

reaching the second polarizer is vertically polarized and so intensity is reduced to

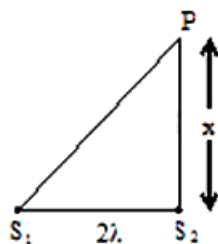
$$I'' = I' (\cos 30^\circ)^2 = \left(\frac{3}{4}\right) \left(\frac{I_0}{2}\right) = \frac{3I_0}{8}$$

Therefore, transmitted light is plane polarised to the vertical at 30° and having intensity $\frac{3I_0}{8}$

10. Two coherent point sources S_1 and S_2 vibrating in phase emit light of wavelength λ . The separation between the sources is 2λ . The smallest distance from S_2 on a line passing through S_2 and perpendicular to S_1S_2 where a minimum intensity occurs is

A) $\frac{7\lambda}{12}$ B) $\frac{15\lambda}{4}$ C) $\frac{\lambda}{2}$ D) $\frac{3\lambda}{4}$

Solution : (Correct Answer: A)



$$S_1P - S_2P = \frac{3\lambda}{2}$$

$$\text{or } \sqrt{4\lambda^2 + x^2} - x = \frac{3\lambda}{2}$$

On solving

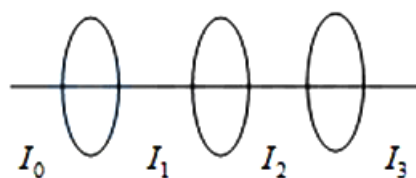
$$x = \frac{7\lambda}{12}$$

11. Unpolarized light of intensity 32 W m^{-2} passes through three polarizers such that transmission axes of the first and second polarizer make an angle 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of the final emerging light will be

A) 32 W m^{-2} B) 3 W m^{-2}
C) 8 W m^{-2} D) 4 W m^{-2}

Solution : (Correct Answer: B)

$$I_0 = 32$$



For 1st polaroid

$$I_1 = \frac{I_0}{2}$$

For 2nd polaroid

$$I_2 = I_1 \cos^2 \theta$$

$$I_2 = \frac{I_0}{2} \times \cos^2 30^\circ$$

$$I_2 = \frac{I_0}{2} \times \frac{3}{4}$$

$$I_2 = \frac{3}{8} I_0$$

As last crosses with 1st polarizer, hence angle between 2nd & last polariser is 60°

$$I_3 = I_2 \cos^2 60^\circ$$

$$I_3 = I_2 \times \frac{1}{4}$$

$$I_3 = \frac{3}{8} I_0 \times \frac{1}{4}$$

$$I_3 = \frac{3I_0}{32}$$

$$I_3 = \frac{3}{32} \times 32$$

$$I_3 = 3 \text{ W m}^{-2}$$

12. If yellow light emitted by sodium lamp in Young's double-slit experiment is replaced by a monochromatic blue light of the same intensity

- A) fringe width will decrease
B) fringe width will increase
C) fringe width will remain unchanged
D) fringe width will become less intense

Solution : (Correct Answer: A)

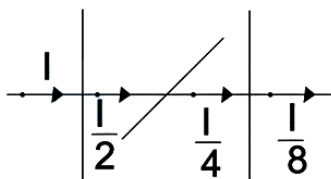
As $\beta = \frac{\lambda D}{d}$ and $\lambda_b < \lambda_y$

$\therefore$ Fringe width β will decrease

13. Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is

- A) 60° B) 0° C) 30° D) 45°

Solution : (Correct Answer: D)



Transformation axis A and B are parallel. If transformation of axis of C is making angle θ with A, it will make angle $90 - \theta$ with B.

$$\frac{I}{2} \cos^2 \theta \cos^2 (90 - \theta) = \frac{I}{8}$$

$$\frac{1}{4} \sin^2 2\theta = \frac{1}{4}$$

$$2\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{4}$$

14. In Young's double-slit experiment, in an interference pattern, the second minimum is observed exactly in front of one slit. The distance between the slits is d and the distance between source and screen is D . The wavelength of the light source used is

- A) $\frac{d^2}{D}$ B) $\frac{d^2}{2D}$ C) $\frac{d^2}{3D}$ D) $\frac{d^2}{4D}$

Solution : (Correct Answer: C)

Distance of second minima from central point is $y = \frac{d}{2}$.

We know that minima is given by,

$$d \sin \theta = \left(n - \frac{1}{2}\right) \lambda$$

$$\frac{dy}{D} = \left(2 - \frac{1}{2}\right) \lambda$$

$$\frac{d}{2} = \frac{D}{d} \left(\frac{2 \times 2 - 1}{2}\right) \lambda$$

$$\therefore \frac{d}{2} = \frac{D}{d} \times \frac{3}{2} \lambda$$

$$\therefore \lambda = \frac{d^2}{3D}$$

15. In Young's double-slit experiment, the separation between the slits is halved and the distance between the slits and the screen is doubled. The fringe width will

- A) remain the same B) be halved
C) be doubled D) be quadrupled

Solution : (Correct Answer: D)

The fringe width in Young's double-slit experiment

$$\beta = \frac{\lambda D}{d}$$

where λ is the wavelength of light used

D is the distance between slit and screen.

d is the distance between the slit.

$$\therefore \beta' = \frac{\lambda(2D)}{\frac{d}{2}} = 4 \frac{\lambda D}{d} = 4\beta$$

16. In Young's double-slit experiment, the number of maxima that are obtained on a screen (including the central maxima), if $\lambda = 2000 \text{ \AA}$ and $d = 7000 \text{ \AA}$ is

- A) 12 B) 7 C) 18 D) 6

Solution : (Correct Answer: B)

For maxima,

$$\Delta x = d \sin \theta = n\lambda$$

$$\therefore \sin \theta = \frac{n\lambda}{d} = \frac{n \times 2000}{7000}$$

Since $\sin \theta$ cannot be greater than 1.

Hence, $n = 0, 1, 2, 3$ only

$\therefore$ 3 maxima on both side of screen and 1 central maxima will be obtained.

$\therefore$ Total maxima = 7

17. What happens to the fringe pattern if in the path of one of the slits a glass plate which absorbs 50% energy is introduced?

- A) The bright fringes become brighter and dark fringes become darker
B) No fringes are observed
C) The fringe width decreases
D) None of the above

Solution : (Correct Answer: D)

Intensity of dark and bright fringes is given by

$$I_{\text{Dark}} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$I_{\text{Bright}} = (\sqrt{I_1} + \sqrt{I_2})^2$$

When there is no glass plate

$$I_1 = I_2 = I_0$$

$$\therefore I_{\text{Dark}} = 0 \text{ and } I_{\text{Bright}} = 4I_0$$

But when glass plate is introduced in the path of one of the slits,

$$I_2 = \frac{I_0}{2}$$

$$\therefore I_{\text{Dark}} \neq 0 \text{ increases and } I_{\text{Bright}} \text{ decreases.}$$

Fringe width remains same.

18. A telescope of objective lens diameter 2 m uses light of wavelength $5000 \text{ \AA}$ for viewing stars. The minimum angular separation between two stars whose image is just resolved by the telescope is

- A) $4 \times 10^{-4} \text{ rad}$ B) $0.25 \times 10^{-6} \text{ rad}$
 C) $0.31 \times 10^{-6} \text{ rad}$ D) $5 \times 10^{-3} \text{ rad}$

Solution : (Correct Answer: C)

Minimum angular separation between two stars whose image is just resolved by telescope is

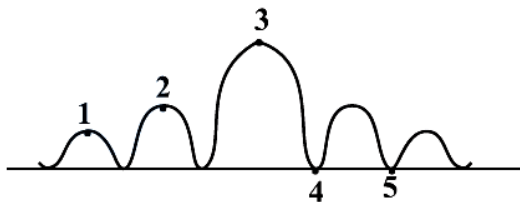
$$\Delta\theta = \frac{1.22\lambda}{a}$$

$$\therefore \Delta\theta = \frac{1.22 \times 5000 \times 10^{-10}}{2}$$

$$\Delta\theta = 3050 \times 10^{-10} = 3.05 \times 10^{-7} \text{ rad}$$

$$\Delta\theta \approx 0.31 \times 10^{-6} \text{ rad}$$

19. The diffraction pattern of a single slit is shown in the figure. The point at which the path difference of the extreme rays is two times the wavelength is



- A) point 1 B) point 2 C) point 4 D) point 5

Solution : (Correct Answer: D)

In diffraction of light at a single slit, the pattern obtained consists of a central bright band having alternate dark and bright bands of decreasing intensity on both sides.

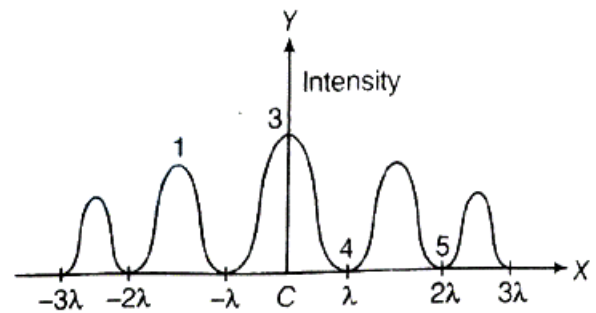
For n^{th} minima (secondary),

$$\sin \theta_n = \frac{n\lambda}{a}$$

For n^{th} maxima (secondary),

$$\sin \theta_n = (2n + 1) \frac{\lambda}{2a}$$

Thus, point 5 corresponds to the path difference of extreme rays equal to 2λ .



20. In single slit diffraction of light of wavelength λ by a slit of width e , the size of the central maximum on a screen at a distance b is

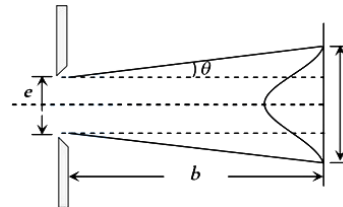
- A) $2b\lambda + e$ B) $\frac{2b\lambda}{e}$
 C) $\frac{2b\lambda}{e} + e$ D) $\frac{2b\lambda}{e} - e$

Solution : (Correct Answer: C)

The direction in which the first minima occurs is θ (say).

$$\text{Then } e \sin \theta = \lambda \text{ or } e\theta = \lambda$$

$$\theta = \frac{\lambda}{e} \quad \therefore \sin \theta = \theta \text{ when } \theta \text{ is small}$$



$$\text{Width of the central maximum} = 2b\theta + e = 2b\frac{\lambda}{e} + e$$

21. In a system of two polarisers, it is found that the intensity of light from the second polarizer is half from that of the first polariser. The angle between their pass axes is

- A) 60° B) 30° C) 0° D) 45°

Solution : (Correct Answer: D)

According to Malus law,

$$I = I_0 \cos^2 \theta$$

$$\text{Here, } I = \frac{I_0}{2}$$

$$\text{Now, } \frac{I_0}{2} = I_0 \cos^2 \theta$$

$$\Rightarrow \cos^2 \theta = \frac{1}{2}$$

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \cos^{-1} \frac{1}{\sqrt{2}} = 45^\circ$$

22. In Young's double slits experiment, if the separation between the slits is halved, and the distance between the slits and the screen is doubled, then the fringe width compared to the original one will be

- A) Unchanged B) Halved C) Doubled D) Quadrupled

Solution : (Correct Answer: D)

The fringe width,

$$\beta = \frac{\lambda D}{d}$$

Where, λ = wavelength of light

d = distance of slit

D = distance of screen from slit.

According to question,

$$d' = \frac{d}{2}$$

$$D = 2D$$

$$\beta' = ?$$

$$\beta' = \frac{\lambda D'}{d'}$$

$$= \frac{\lambda 2D}{d/2} = 4\beta$$

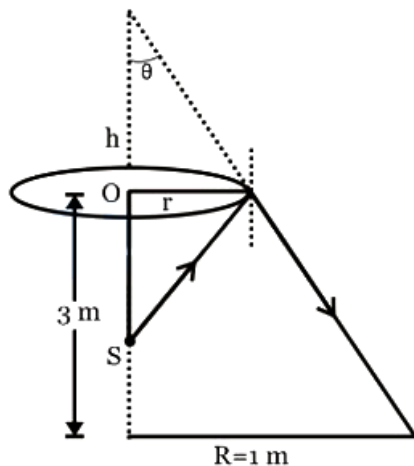
So, the fringe width will be quadrupled.

23. A plane mirror of circular shape with radius $r = 20$ cm is fixed to the ceiling. A bulb is to be placed on the axis of the mirror. A circular area of the radius $R = 1$ m on the floor is to be illuminated after reflection of light from the mirror. The height of the room is 3 m. What is the maximum distance from the centre of the mirror and the bulb so that the required area is illuminated?

- A) 75 cm B) 70 cm
C) 80 cm D) 100 cm

Solution : (Correct Answer: A)

$$\tan \theta = \frac{r}{h} = \frac{R}{3+h}$$



$$\frac{0.20}{h} = \frac{1}{3+h} \Rightarrow 0.6 + 0.2h = h$$

$$0.8h = 0.6 \Rightarrow h = \frac{3}{4} \text{ m}$$

$$h = 75 \text{ cm}$$

24. The frequency of a light wave in a material is 2×10^{14} Hz and wavelength is $5000 \text{ \AA}$. The refractive index of material will be

- A) 1.40 B) 1.50 C) 3.00 D) 1.33

Solution : (Correct Answer: C)

The velocity of light water in the material is

$$v = v\lambda \quad \dots (i)$$

Refractive index of the material is

$$\mu = \frac{c}{v} \quad \dots (ii)$$

Where c is the speed of light in vacuum or air

$$\text{or } \mu = \frac{c}{v\lambda} \quad \dots (iii)$$

$$\text{Given, } v = 2 \times 10^{14} \text{ Hz,}$$

$$\lambda = 5000 \text{ \AA} = 5000 \times 10^{-10} \text{ m}$$

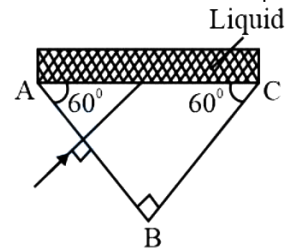
$$c = 3 \times 10^8 \text{ m s}^{-1}$$

Hence, from Eq (iii), we get

$$\mu = \frac{3 \times 10^8}{2 \times 10^{14} \times 5000 \times 10^{-10}} = 3.00$$

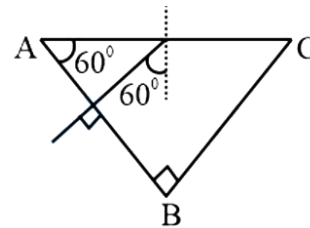
25.

Light is incident normally on the face AB of a prism as shown in the figure. A layer of liquid of refractive index μ is applied on the face AC of the prism. If the prism is made of glass of refractive index $\frac{3}{2}$, then the limit of μ for which total internal reflection takes place on face AC is



- A) $\mu < \frac{3\sqrt{3}}{4}$ B) $\mu > \sqrt{\frac{5}{2}}$ C) $\mu > \sqrt{2}$ D) $\mu < \sqrt{3}$

Solution : (Correct Answer: A)



For critical angle at the glass-liquid interface we have

$$\mu_1 \sin C = \mu_2 \sin 90^\circ$$

$$\Rightarrow \frac{3}{2} \sin C = \mu$$

$$\Rightarrow \sin C = \frac{2\mu}{3}$$

For total internal reflection

$$60^\circ > C$$

$$\Rightarrow \sin 60^\circ > \sin C$$

$$\Rightarrow \frac{\sqrt{3}}{2} > \frac{2\mu}{3}$$

$$\mu < \frac{3\sqrt{3}}{4}$$

26. Laser light has the following property

- A) Laser light always lies in the X-ray region
B) Laser light does not have directionality property
C) Laser light is white light
D) Laser light is highly coherent

Solution : (Correct Answer: D)

Laser is a process by which we get a light beam which is coherent, highly monochromatic and almost perfectly parallel. A beam of laser can go to the Moon and return to the Earth without much loss of intensity.

27. Two lenses of power +10 D and -5 D are placed in contact. Where should an object be held from the lens, so as to obtain a virtual image of magnification 2?

A) 5 cm B) -5 cm
C) 10 cm D) -10 cm

Solution : (Correct Answer: D)

Here, $P_1 = +10$ D, $P_2 = -5$ D

Combined power,

$$P = P_1 + P_2 = 10 \text{ D} - 5 \text{ D}$$

$$P = 5 \text{ D}$$

$\therefore$ Combined focal length

$$= \frac{1}{P} \text{ m} = \frac{100}{P} \text{ cm} = \frac{100}{5} = 20 \text{ cm}$$

Given magnification,

$$m = 2$$

$$\frac{v}{u} = 2$$

$$v = 2u$$

By lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{2u} - \frac{1}{u} = \frac{1}{20}$$

$$\frac{1-2}{2u} = \frac{1}{20}$$

$$2u = 20$$

$$u = -10 \text{ cm}$$

Hence, to obtain virtual image, object should be place at a distance of 10 cm.

28. A convex, rearview mirror of focal length 20 cm, is fitted in a car. A second car 2 m broad and 1.6 m high is 6 m away from the first car and overtakes the first car at a relative speed of 15 m s^{-1} , then the speed of the image of the second car with respect to the first car is

A) 0.016 m s^{-1}
B) 0.0257 m s^{-1}
C) 0.162 m s^{-1}
D) 0.0073 m s^{-1}

Solution : (Correct Answer: A)

Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Differentiating w.r.t. t,

$$0 = -\frac{1}{v^2} \frac{dv}{dt} - \frac{1}{u^2} \frac{du}{dt}$$

$$\frac{dv}{dt} = -\frac{v^2}{u^2} \frac{du}{dt}$$

$$\text{But } \frac{du}{dt} = 15 \text{ m s}^{-1}$$

$$\text{Now } \frac{1}{v} = \left(\frac{1}{20} + \frac{1}{600} \right) \therefore v = \frac{600}{31} \text{ cm}$$

$$\text{and } \frac{v}{u} = \frac{600/31}{600} = \frac{1}{31}$$

$$\text{Speed of image} = \frac{dv}{dt} = -\left(\frac{1}{31}\right)^2 \times 15$$

$$= 0.016 \text{ m s}^{-1}$$

29. A ray of light strikes a transparent slab of the refractive index of $\sqrt{2}$ at an angle of incidence 45° . The angle between the reflected and refracted rays is

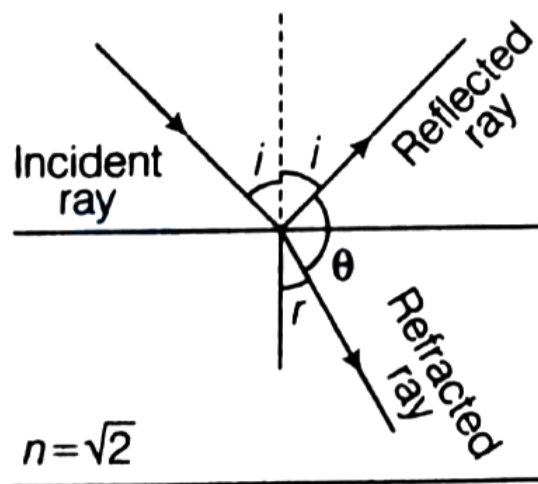
A) 95° B) 120°
C) 135° D) 105°

Solution : (Correct Answer: D)

According to the figure,

Given, $i = 45^\circ$

Let the angle between the reflected ray and refracted ray is



Apply Snell's law at the air-glass surface,

$$\frac{\sin i}{\sin r} = \frac{n}{1}$$

$$\Rightarrow \frac{\sin 45^\circ}{\sin r} = \sqrt{2}$$

$$\sin r = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}$$

$$\therefore r = 30^\circ$$

From the diagram, $\theta = 180 - (i + r)$

$$= 180 - (45^\circ + 30^\circ)$$

$$= 180^\circ - 75^\circ = 105^\circ$$

30. The red light of a wavelength $6400 \text{ \AA}$ in the air has a wavelength of $4000 \text{ \AA}$ in glass. If the wavelength of violet light in air is $4400 \text{ \AA}$, then the wavelength in the glass is

A) $2570 \text{ \AA}$ B) $2750 \text{ \AA}$
C) $1600 \text{ \AA}$ D) $2510 \text{ \AA}$

Solution : (Correct Answer: B)

$$\mu_g = \frac{(\lambda_v)_{\text{air}}}{(\lambda_v)_{\text{glass}}}$$

$$\text{Now, } \mu_g = \frac{(\lambda_v)_{\text{air}}}{(\lambda_v)_{\text{glass}}} = \frac{6400}{4000}$$

$$(\lambda_v)_{\text{glass}} = \frac{(\lambda_v)_{\text{air}}}{\mu_g} = \frac{4400}{1.6} = 2750 \text{ \AA}$$

31. Magnification of a compound microscope is 30. The focal length of eye-piece is 5 cm and the image is formed at a distance of 'distinct vision' of 25 cm. Magnification of the objective lens is

A) 6 B) 5 C) 7.5 D) 10

Solution : (Correct Answer: B)

$$m = m_1 \cdot m_2 = 30$$

$$\text{for eye-piece } \frac{1}{v_1} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{-1}{25} - \frac{1}{u} = \frac{1}{5}$$

$$\therefore u = \frac{-25}{6}$$

$$\therefore m_2 = \frac{-25 \times 6}{-25} = 6$$

$$\therefore m_1 = 5$$

32. For a prism of refractive index, 1.732, the angle of minimum deviation is equal to the angle of prism. Then the angle of the prism is

A) 60° B) 70°
C) 50° D) none of these

Solution : (Correct Answer: A)

$$\text{We know } \mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin \frac{A}{2}}$$

$$A = \delta_{\min}$$

$$\therefore 1.732 = \frac{\sin \frac{A}{2}}{\sin \frac{A}{2}}$$

$$\therefore A = 60^\circ$$

33. Near and far points of human eye are

A) 25 cm and infinite
B) 25 cm and 100 cm
C) 55 cm and 200 cm
D) 0 cm and 25 cm

Solution : (Correct Answer: A)

Our eye lens has a power to adjust its focal length to see the nearer and farther objects, this process of adjusting focal length is called accommodation. However, if the object is brought too close or brought too far from the eye, the focal length cannot be adjusted to form the image on the retina. Thus, there is minimum or maximum distance for the clear vision of an object. For a normal eye, near point or least distant vision $D = 25$ cm and far point $= \infty$

34. In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm. An object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens. The distance between the two lenses is

A) 6.00 cm B) 7.75 cm C) 9.25 cm D) 11.00 cm

Solution : (Correct Answer: D)

$$f_o = 1.5 \text{ cm}, f_e = 6.25 \text{ cm}, u_o = -D = -25 \text{ cm}$$

$$\text{By objective lens } \frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$$

$$\frac{1}{1.5} = \frac{1}{v_o} - \frac{1}{-25}$$

$$\Rightarrow \frac{1}{v_o} = \frac{1}{1.5} - \frac{1}{25}$$

$$\text{or } v_o = 6 \text{ cm}$$

By eyepiece

$$\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e}$$

$$\frac{1}{6.25} = \frac{1}{-25} - \frac{1}{-u_e}$$

$$\Rightarrow \frac{1}{u_e} = \frac{1}{6.25} + \frac{1}{25}$$

$$\Rightarrow \frac{1}{u_e} = \frac{4}{25} + \frac{1}{25} = \frac{5}{25}$$

$$\Rightarrow \frac{1}{u_e} = \frac{1}{5}$$

$$u_e = 5 \text{ cm}, \text{ Length of tube } = L = v_o + u_e = 6.0 + 5.0 = 11 \text{ cm}$$

35. In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm and an object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens. The distance between the two lenses is

A) 6.00 cm B) 7.75 cm C) 9.25 cm D) 11.00 cm

Solution : (Correct Answer: D)

$$f_o = 1.5 \text{ cm}, f_e = 6.25 \text{ cm}, u_o = -2 \text{ cm}, v_e = -D = -25 \text{ cm}$$

$$\text{By objective lens } \frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$$

$$\frac{1}{1.5} = \frac{1}{v_o} - \frac{1}{-2}$$

$$\Rightarrow \frac{1}{v_o} = \frac{1}{1.5} - \frac{1}{2} \text{ or } v_o = 6 \text{ cm}$$

$$\text{By eyepiece } \frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e}$$

$$\frac{1}{6.25} = \frac{1}{-25} - \frac{1}{-u_e}$$

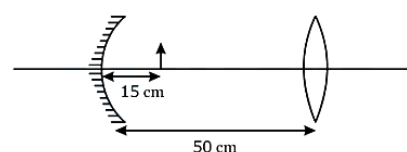
$$\Rightarrow \frac{1}{u_e} = \frac{1}{6.25} + \frac{1}{25} = \frac{4}{25} + \frac{1}{25} = \frac{5}{25}$$

$$u_e = 5 \text{ cm}$$

$$\text{Length of tube } = L = v_o + u_e = 6.0 \text{ cm} + 5.0 \text{ cm}$$

$$L = 11 \text{ cm}$$

36. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1), as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image, formed by this combination, has magnification M_1 . When the set-up is kept in a medium of refractive index $\frac{7}{6}$, the magnification becomes M_2 . The magnitude $\left| \frac{M_2}{M_1} \right|$ is



A) 7 B) 5 C) 3 D) 6

Solution : (Correct Answer: A)

For reflection from a concave mirror,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{15} = \frac{-1}{10}$$

$$\frac{1}{v} = \frac{1}{15} - \frac{1}{10} = -\frac{1}{30}$$

$$\therefore v = -30$$

$$\text{Magnification } (m_1) = -\frac{v}{u} = -2$$

Now for refraction from lens,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{20} = \frac{1}{20}$$

$$\therefore \text{Magnification } (m_2) = \frac{v}{u} = -1$$

$$\therefore M_1 = m_1 m_2 = 2$$

Now when the set-up is immersed in liquid, no effect for the image formed by mirror.

$$\text{We have } (\mu_L - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{10}$$

$$\Rightarrow \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{5}$$

When lens is immersed in liquid,

$$\frac{1}{f_{\text{lens}}} = \left(\frac{\mu_L}{\mu_S} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{2}{7} \times \frac{1}{5} = \frac{2}{35}$$

$$\therefore \frac{1}{v} - \frac{1}{u} = \frac{1}{f_{\text{liquid}}}$$

$$\Rightarrow \frac{1}{v} = \frac{2}{35} - \frac{1}{20} = \frac{8-7}{140} = \frac{1}{140}$$

$$\therefore \text{Magnification} = -\frac{140}{20} = -7$$

$$\therefore M_2 = 2 \times 7 = 14$$

$$\therefore \left| \frac{M_2}{M_1} \right| = 7$$

37. An equilateral prism deviates a ray through 45° for two angles of incidence differing by 20° . The angles of incidence are :

- A) 60° and 40°
 B) 50° and 70°
 C) $62^\circ 30'$ and $42^\circ 30'$
 D) $60^\circ 30'$ and $40^\circ 30'$

Solution : (Correct Answer: C)

$$\delta = i_1 + i_2 - A$$

$$45^\circ = i_1 + i_2 - 60^\circ$$

$$i_1 + i_2 = 105^\circ$$

$$i_1 - i_2 = 20^\circ$$

$$\text{or } 2i_1 = 125^\circ$$

$$i_1 = 62^\circ 30', \quad i_2 = 42^\circ 30'$$

38. If the angle of minimum deviation is of 60° for an equilateral prism, then the refractive index of the material of the prism is

- A) 1.41 B) 1.5 C) 1.6 D) 1.73

Solution : (Correct Answer: D)

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin \frac{A}{2}}$$

For equilateral prism, $\angle A = 60^\circ$

$$\mu = \frac{\sin\left(\frac{60^\circ+60^\circ}{2}\right)}{\sin \frac{60^\circ}{2}} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3} = 1.73$$

39. Time taken by the sunlight to pass through a slab of 4 cm and refractive index 1.5 is ____ s.

- A) 2×10^{10} B) 2×10^{-8} C) 2×10^8 D) 2×10^{-10}

Solution : (Correct Answer: D)

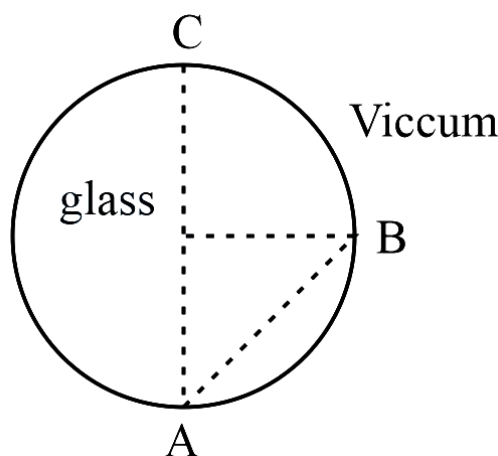
$$d = 4 \text{ cm}$$

$$\mu = 1.5$$

$$\mu = \frac{c}{v} \Rightarrow v = \frac{c}{\mu} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m s}^{-1}$$

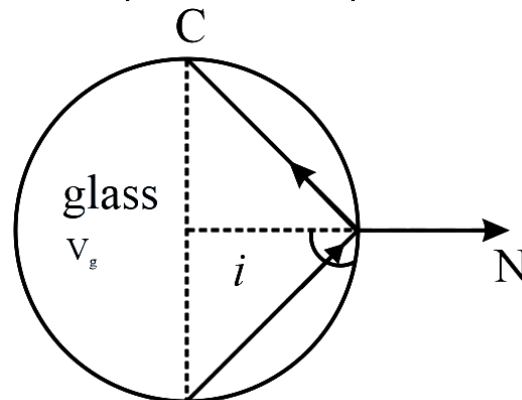
$$\text{Time } t = \frac{d}{v} = \frac{4 \times 10^{-2}}{2 \times 10^8} = 2 \times 10^{-10} \text{ s}$$

40. It is found that all electromagnetic signals sent from A towards B reach point C. The speed of electromagnetic signals in glass cannot be



- A) $1.0 \times 10^8 \text{ m s}^{-1}$
 B) $2.15 \times 10^8 \text{ m s}^{-1}$
 C) $2 \times 10^7 \text{ m s}^{-1}$
 D) $4 \times 10^7 \text{ m s}^{-1}$

Solution : (Correct Answer: B)



There TIR at B,

$$i > i_C$$

$$\therefore 45^\circ > \sin^{-1} \left(\frac{1}{\mu} \right)$$

$$\therefore 45^\circ > \sin^{-1} \left(\frac{v_g}{c} \right)$$

$$\therefore \sin 45^\circ > \frac{v_g}{c}$$

$$\therefore v_g < c \times \sin 45^\circ$$

$$\therefore v_g < 3 \times 10^8 \times \frac{1}{\sqrt{2}}$$

$$\therefore v_g < 2.14 \times 10^8 \text{ m s}^{-1}$$

41. The refractive index of water is 1.33. What will be the speed of light in water?

- A) $3 \times 10^8 \text{ ms}^{-1}$
 B) $4 \times 10^8 \text{ ms}^{-1}$
 C) $1.33 \times 10^8 \text{ ms}^{-1}$
 D) $2.26 \times 10^8 \text{ ms}^{-1}$

Solution : (Correct Answer: D)

We know that refractive index of a medium is given by :

$$\mu = \frac{c}{v}$$

$$\therefore v = \frac{c}{\mu} = \frac{3 \times 10^8}{1.33} = 2.26 \times 10^8 \text{ ms}^{-1}$$

42. A ray of light passing through a prism of refractive index $\sqrt{2}$ undergoes minimum deviation. It is found that the angle of incidence is double the angle of refraction within the prism. The angle of the prism is

- A) 60° B) 90° C) 75° D) 30°

Solution : (Correct Answer: B)

According to given problem, $i = 2r = A$

$$(\because \text{angle of prism, } A = 2r)$$

$$\text{Since, } \delta_m = 2i - A$$

$$\Rightarrow \delta_m = 2A - A = A$$

$$\text{We have, } n = \frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})} \Rightarrow \sqrt{2} = \frac{\sin A}{\sin(\frac{A}{2})}$$

$$\Rightarrow \sqrt{2} = \frac{2 \sin(\frac{A}{2}) \cos(\frac{A}{2})}{\sin(\frac{A}{2})}$$

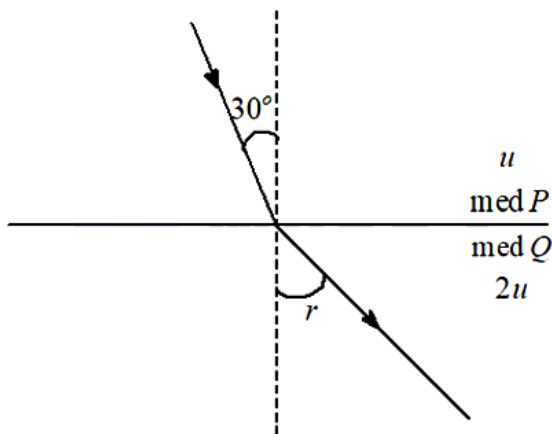
$$\Rightarrow \cos\left(\frac{A}{2}\right) = \frac{\sqrt{2}}{2} \Rightarrow \frac{A}{2} = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\Rightarrow \frac{A}{2} = 45^\circ \Rightarrow A = 90^\circ$$

43. A wave has velocity u in medium P and velocity $2u$ in medium Q . If the wave is refracted from medium P to medium Q at an angle of incidence 30° , then the angle of refraction will be

- A) 30° B) 45° C) 60° D) 90°

Solution : (Correct Answer: D)



From Snell's law,

$$\mu_P \sin 30^\circ = \mu_Q \sin r$$

$$\sin r = \frac{\mu_P}{\mu_Q} \sin 30^\circ$$

$$\text{as } \mu \propto \frac{1}{v}$$

$$\sin r = \frac{2u}{u} \frac{1}{2} = 1$$

$$r = 90^\circ$$

44. A double convex thin lens made of the refractive index 1.6 has radii of curvature 15 cm each. The focal length of this lens when immersed in a fluid of refractive index 1.63, is

- A) 25 cm B) 125 cm
 C) 250 cm D) -407.5 cm

Solution : (Correct Answer: D)

Focal length of lens is given by

$$\frac{1}{f_w} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$= \left(\frac{1.6}{1.63} - 1 \right) \left(\frac{1}{15} + \frac{1}{15} \right)$$

$$= -\frac{0.03 \times 2}{1.63 \times 15} = -\frac{.06}{1.63 \times 15}$$

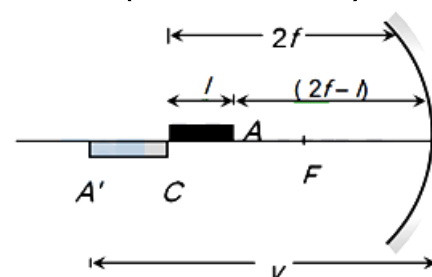
$$\text{so, } f_w = -\frac{815}{2}$$

$$= -407.5 \text{ cm}$$

45. A thin rod of 5 cm length is kept along the axis of a concave mirror of 10 cm focal length such that its image is real and magnified and one end touches the rod. Its magnification will be

- A) 1 B) 2 C) 3 D) 4

Solution : (Correct Answer: B)



End A of the rod acts as an object for mirror and A' will be its image so $u = 2f - l = 20 - 5 = 15 \text{ cm}$

$$\therefore \frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{-10} = \frac{1}{v} - \frac{1}{15} \Rightarrow v = -30 \text{ cm}.$$

$$\text{Now } m = \frac{\text{Length of image}}{\text{Length of object}} = \frac{(30-20)}{5} = 2$$

Chemistry

46. Which of the following geminal diols is most unstable:

- A) $\text{F}_3\text{C}-\text{C}(\text{OH})_2$ B) $\text{H}_3\text{C}-\text{C}(\text{OH})_2$ C) $\text{ClCH}_2-\text{C}(\text{OH})_2$ D) $\text{Cl}_3\text{C}-\text{C}(\text{OH})_2$

Solution : (Correct Answer: B)

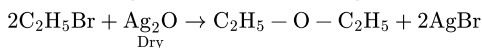
Due to +I effect of $-\text{CH}_3$ group, electron density on oxygen increases and thus repulsion increases.

47. The compound formed when Ethyl bromide is heated with dry Silver oxide is

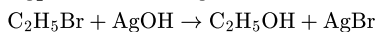
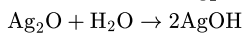
- A) dimethylether B) diethylether

C) methylalcohol D) ethylalcohol

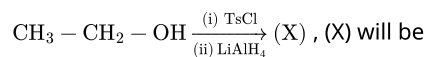
Solution : (Correct Answer: B)



If we take moist Ag_2O then alcohol is formed



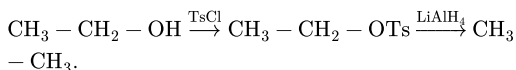
48. In the reaction



- A) $\text{CH}_2 = \text{CH}_2$ B) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$
 C) $\text{CH}_3 - \text{CH}_3$ D) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$

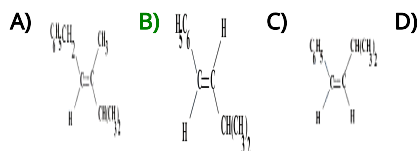
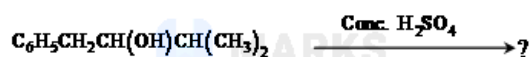
Solution : (Correct Answer: C)

Here the compound X is ethane and formed as follows



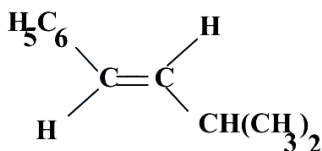
Hint : It involves substitution followed by reduction.

49. The main product of the following reactions is



Solution : (Correct Answer: B)

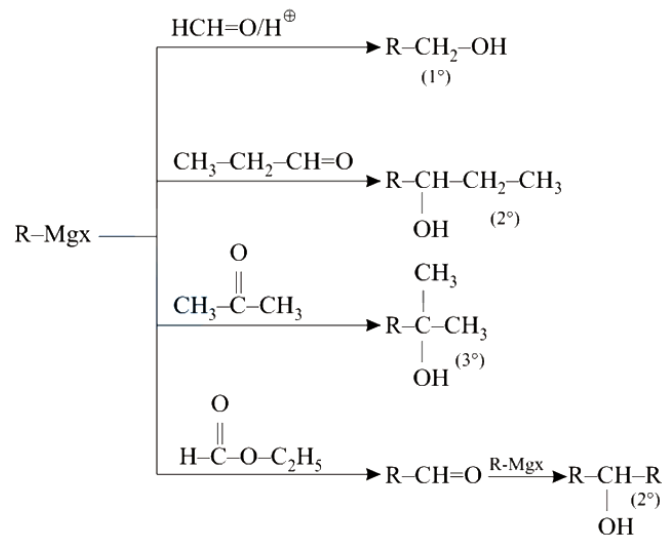
Dehydration can produce two different alkenes, major product is formed according to **Saytzeff rule** i.e. more substituted alkene (alkene having lesser number of hydrogen atoms on the two doubly bonded carbon atoms) is the major product. E_1 elimination takes place which is anti elimination.



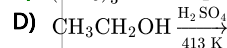
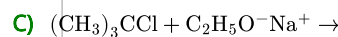
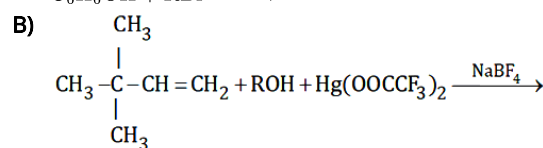
50. Which of the compounds HCHO (I), $\text{CH}_3\text{CH}_2\text{CHO}$ (II), CH_3COCH_3 (III) and HCOOC_2H_5 (IV) will give a secondary alcohol on reaction with excess Grignard reagent, followed by hydrolysis? Select the correct answer using the codes given below

- A) II only B) III only C) I and IV D) II and IV

Solution : (Correct Answer: D)

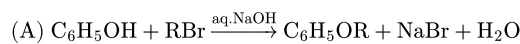
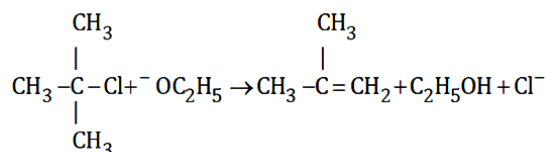


51. Which of the following reaction does not form an ether?

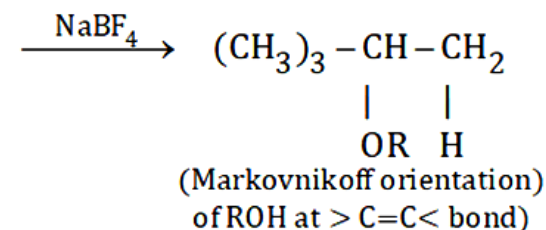
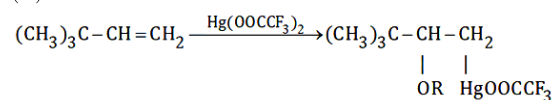


Solution : (Correct Answer: C)

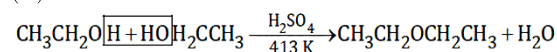
Because of strong basicity of the alkoxide ion, elimination reaction predominates over substitution reaction, if we use tertiary halide.



(B)



(D)



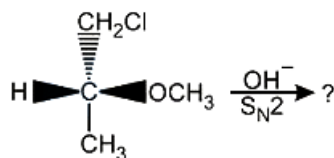
52. Which of the following can work as a dehydrating agent for alcohols

- A) H_2SO_4 B) Al_2O_3 C) H_3PO_4 D) All of these

Solution : (Correct Answer: D)

H_2SO_4 , Al_2O_3 and H_3PO_4 all can act as dehydrating agent.

53. What is the major product of the reaction ?



- A) B) C) D)

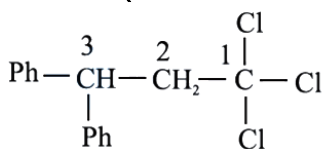
Solution : (Correct Answer: B)

α -carbon is achiral, hence retention of configuration will be observed at chiral β -carbon.

Configuration by S_N2 substitution changes at α , C atom but β , C atom is chiral so configuration to be retained at β , C atom.

54.
$$\begin{array}{c} \text{C}_6\text{H}_5 - \text{CH} - \text{CH}_2 - \text{CCl}_3 \\ | \\ \text{C}_6\text{H}_5 \end{array}$$
 The IUPAC name is
- A) 1, 1, 1- trichloro -3, 3- diphenyl propane
B) 1, 1- diphenyl -3, 3, 3- trichloro propane
C) Both (A) and (B)
D) None of these

Solution : (Correct Answer: A)



55. In the given reaction

$$\text{R} - \text{I} + \text{NaCN} \xrightarrow{\text{DMF}} \text{R} - \text{C} \equiv \text{N}$$

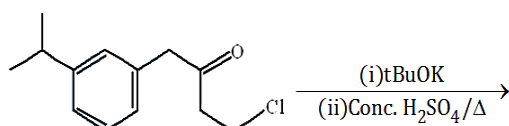
 $\text{R} - \text{I}$ may not be
- A) Primary B) Secondary C) Tertiary D) $\text{CH}_3 - \text{I}$

Solution : (Correct Answer: C)

$$\text{R} - \text{I} + \text{NaCN} \xrightarrow{\text{DMF}} \text{R} - \text{C} \equiv \text{N}$$

 Here $\text{R} - \text{I}$ cannot be tertiary halide as it is S_N2 reaction.

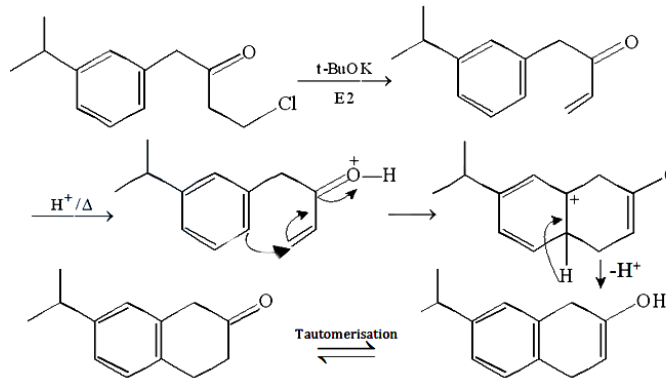
56. The major product of the following reaction is



- A) B) C) D)

Solution : (Correct Answer: D)

This reaction occurs by the following mechanism given here.



57. Which of the aromatic compounds react fastest with methoxide ion?

- A) B) C) D)

Solution : (Correct Answer: B)

A nitro from para position activate more for S_NAr reaction due to electron withdrawing resonance effect than two nitro from meta position by inductive effect (-I).

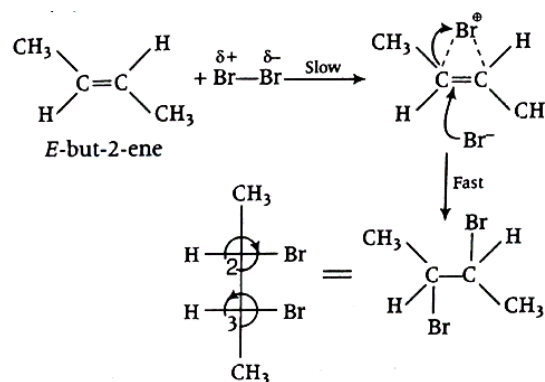
58. The addition of Br_2 to (E)-but-2-ene gives

- A) (R, R) - 2 - 3- dibromobutane
B) (S, R) - 2 - 3- dibromobutane
C) (R, S) - 2 - 3- dibromobutane
D) A mixture of (R, R) and (S, S) 2, 3 dibromobutane

Solution : (Correct Answer: C)

The addition of bromine to E - but-2-ene gives (R, S) - 2 - 3- dibromobutane, and it is always anti as bromine atoms added are on opposite sides.

In (E) - but -2- ene addition of Br_2 occurs as:



(R, S) - 2 - 3- dibromobutane.

59. Which of the following will not form Grignard reagent with Mg/dry ether?

- (1) $\text{CH}_2\text{Br} - \text{CH}_2\text{Br}$ (2) $\text{CH}_2\text{Br} - \text{CH}_2 - \text{CH}_2\text{Br}$

- (3) (4)

Select the correct answer using the codes given below

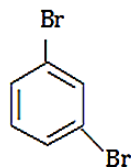
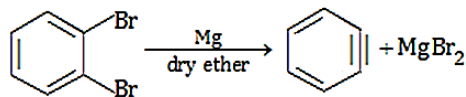
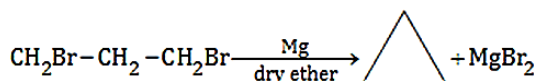
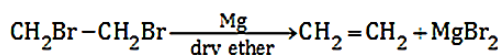
- A) Only (4) B) Only (2)

C) 1, 2 and 3

D) 1, 2, 3 and 4

Solution : (Correct Answer: C)

All these are vicinal dihalide and do not form Grignard reagent as a vic dihalide gives elimination reaction with Mg.



can form Grignard reagent.

60. How is the relative adsorption of each component of the mixture expressed in thin layer chromatography?

- A) By their color
B) By their boiling points
C) By their density
D) By their retardation factor (R_f) value

Solution : (Correct Answer: D)

The relative adsorption of each component of the mixture in thin layer chromatography is expressed in terms of its retardation factor (R_f) value.

61. What is the common suffix for the names of straight chain saturated hydrocarbons?

- A) -ene B) -ane C) -ol D) -ine

Solution : (Correct Answer: B)

The common suffix for the names of straight chain saturated hydrocarbons is '-ane.'

62. What is the relationship between hybridization and bond length?

- A) Hybridization has no effect on bond length.
B) sp hybridization results in shorter bonds than sp^2 and sp^3 hybridization.
C) sp^3 hybridization results in shorter bonds than sp and sp^2 hybridization.
D) sp^2 hybridization results in shorter bonds than sp and sp^3 hybridization.

Solution : (Correct Answer: B)

The sp hybrid orbital forms shorter bonds than the sp^2 and sp^3 hybrid orbitals. (This is due to the greater s character in sp orbitals, making the bonds stronger and therefore shorter.)

63. How does the relative electronegativity of carbon atoms with different hybrid orbitals affect the properties of molecules?

- A) It has no effect on the properties of molecules.
B) It affects the color of the molecules.
C) It influences the physical and chemical properties of the molecules.
D) It changes the odor of the molecules.

Solution : (Correct Answer: C)

The relative electronegativity of carbon atoms with different hybrid orbitals is reflected in several

physical and chemical properties of the molecules.

64. What is necessary for proper sideways overlap in π bond formation?

- A) Perpendicular orientation of p orbitals
B) Parallel orientation of p orbitals
C) s orbital overlap
D) d orbital overlap

Solution : (Correct Answer: B)

In π bond formation, a proper sideways overlap requires the parallel orientation of the two p orbitals on adjacent atoms.

65. Which atoms are specifically written in bond-line structural representations?

- A) Carbon and hydrogen atoms
B) Oxygen, chlorine, and nitrogen atoms
C) Hydrogen atoms only
D) All atoms except carbon and hydrogen

Solution : (Correct Answer: B)

In bond-line structural representations, only oxygen, chlorine, nitrogen, etc., are specifically written, while carbon and hydrogen atoms are not shown.

66. Who synthesized acetic acid in 1845, demonstrating that organic compounds could be synthesized from inorganic sources in a laboratory?

- A) F. Wohler B) Berthel C) Berzilius D) Kolbe

Solution : (Correct Answer: D)

Kolbe synthesized acetic acid in 1845, showing that organic compounds could be synthesized from inorganic sources in a laboratory.

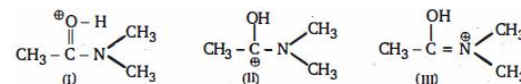
67. Decreasing order of stability of following carbocations is

- (A) $m - \text{CH}_3\text{OPhCH}_2^+$ (B) $p - \text{CH}_3\text{OPhCH}_2^+$
(C) PhCH_2^+ (D) $p - \text{NO}_2\text{PhCH}_2^+$

- A) $A > B > C > D$
B) $B > C > D > A$
C) $C > B > A > D$
D) $B > C > A > D$

Solution : (Correct Answer: D)

68. The correct stability order of the given canonical structures is

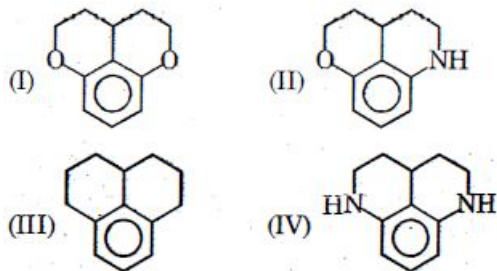


- A) $I > II > III$
B) $III > I > II$
C) $I > III > II$
D) $II > III > I$

Solution : (Correct Answer: B)

(b) Structure III & I has complete octet for all atoms but III is more stable than I because $N^+ > O^+$ is the stability order.

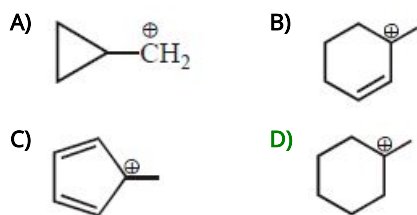
69. In above compounds, decreasing order of rate of electrophilic aromatic substitution will be ?



- A) $I > II > III > IV$
 B) $III > IV > II > I$
 C) $IV > II > I > III$
 D) $IV > I > II > III$

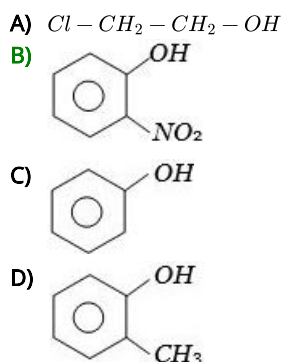
Solution : (Correct Answer: C)

70. Identify the correct order of stability of carbocation



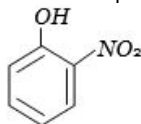
Solution : (Correct Answer: D)

71. Which one of the following compounds is most acidic



Solution : (Correct Answer: B)

(b) Electron withdrawing group increases acidic character due to $-I$ and $-R$ effect of NO_2 hence orthonitrophenol is most acidic.



72. Which behaves both as a nucleophile and electrophile

- A) CH_3NH_2 B) CH_3Cl
 C) CH_3CN D) CH_3OH

Solution : (Correct Answer: C)

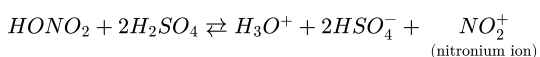
(c) CH_3CN behaves both as nucleophile and electrophile.

73. In the mixture of conc. H_2SO_4 and HNO_3 the nitrating species is

- A) N_2O_4 B) NO_2^+ C) NO_2 D) NO_2^-

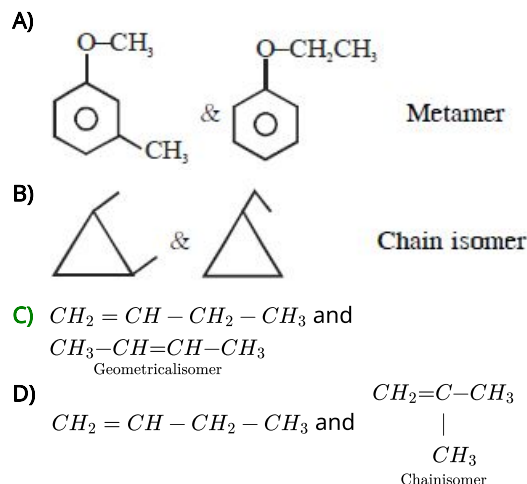
Solution : (Correct Answer: B)

The process of nitration takes place as below



So nitrating species is NO_2^+ ion.

74. Which of the following is not correctly matched



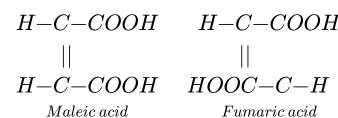
Solution : (Correct Answer: C)

75. Maleic acid and fumaric acids are

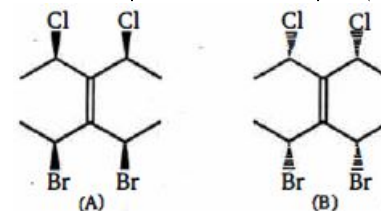
- A) Chain isomers
 B) Functional isomers
 C) Tautomers
 D) Geometrical isomers

Solution : (Correct Answer: D)

Maleic acid and fumaric acids are geometrical isomers.



76. Relationship between above pair (A) & (B) is

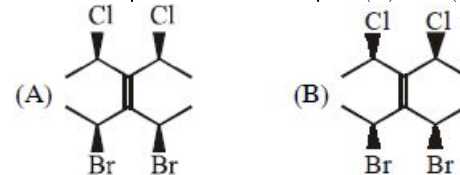


- A) Enantiomer B) Diastereomers
 C) Identical D) Structural isomer

Solution : (Correct Answer: C)

(c) Identical.

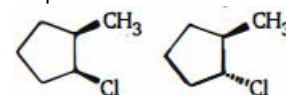
77. Relationship between above pair (A) and (B) is



- A) Enantiomer B) Diastereomers
 C) Identical D) Structural isomer

Solution : (Correct Answer: B)

78. The following compounds are identical with respect to

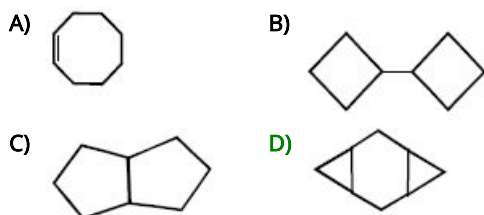


- A) molecular composition
 B) boiling point
 C) melting point
 D) IUPAC name

Solution : (Correct Answer: A)

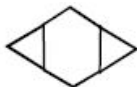
(a) Both molecule has same molecular formula $C_6H_{11}Cl$ so have same molecular wt. and composition because both have substituents on same atoms.

79. Which one of the compound is not isomer of others?



Solution : (Correct Answer: D)

It has different molecular formula and different DU

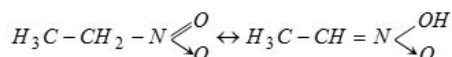


80. Nitroethane can exhibit one of the following kind of isomerism

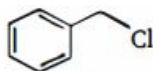
A) Metamerism B) Optical activity
C) Tautomerism D) Position isomerism

Solution : (Correct Answer: C)

(c) Nitroalkanes exhibit tautomerism. In it, $\alpha - H$ -atom is labile and form nitrolic acid.



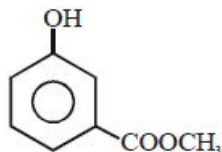
81. The IUPAC name of (figure) is



A) Chloromethylbenzene
B) Chlorophenylmethane
C) (a) and (b) both
D) none of these

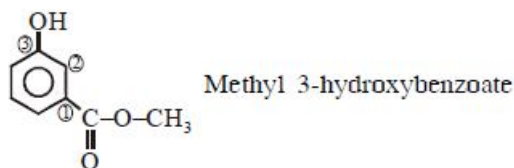
Solution : (Correct Answer: B)

82. The IUPAC name of (Image) is



A) 2-hydroxymethyl benzoate
B) 2-hydroxyethyl benzoate
C) Methyl 3-hydroxy benzoate
D) 1-hydroxymethyl benzoate

Solution : (Correct Answer: C)



83. The incorrect statement regarding ethyne is

A) The C - C bonds in ethyne is shorter than that in ethene
B) Both carbons are sp hybridised
C) Ethyne is linear

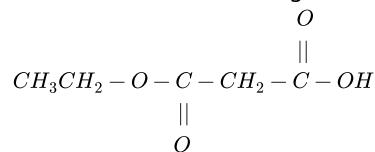
D) The carbon-carbon bonds in ethyne is weaker than that in ethene

Solution : (Correct Answer: D)

The carbon-carbon bonds in ethyne is stronger than that in ethene.

(H - C $\equiv$ C - H) Ethyne is linear and carbon atoms are sp hybridised.

84. The correct IUPAC name of given compound



A) Ethyl carboxylate ethanoic acid
B) Ethoxy carbonyl ethanoic acid
C) Ethyl -2- oxo propanoic acid
D) 3- oxo -3- ethoxy propanoic acid

Solution : (Correct Answer: B)

85. All the following IUPAC names are correct except :

A) 1-chloro -1- ethoxy propane
B) 1- amino -1- ethoxypropane
C) 1- ethoxy -2- propanol
D) 1- ethoxy -1- propanamine

Solution : (Correct Answer: B)

The structure of 1 - Chloro-1-ethoxy - propane will be: $CH_3CH_2CH(Cl)OCH_2CH_3$

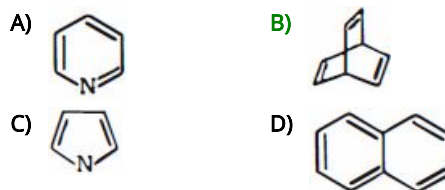
The structure of 1 - Ethoxy - 2 - propanol will be: $CH_3CH(OH)CH_2OCH_2CH_3$

The structure of 1 - Amino -1 - ethoxy - propane will be: $CH_3CH_2CH(NH_2)OCH_2CH_3$

But the IUPAC name 1 - Amino - 1 - ethoxy - propane is incorrect as amino group has higher priority than ethoxy group.

So, all of the IUPAC names are correct except 1 - Amino-1-ethoxy -propane.

86. Which of the following compounds would not be considered aromatic in its behaviour?



Solution : (Correct Answer: B)

(b) Non-planar structure are not aromatic.

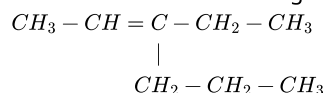
87. In which of the following species is the underlined carbon having sp^3 hybridisation

A) $CH_3\underline{C}OOH$ B) $CH_3\underline{C}H_2OH$
C) $CH_3\underline{C}OCH_3$ D) $CH_2 = \underline{C}H - CH_3$

Solution : (Correct Answer: B)

All the three other options are sp^2 hybridized except CH_3CH_2OH which is sp^3 hybridized.

88. IUPAC name of the following compound will be



A) 3-ethyl -2- hexene
B) 3-propyl -2- hexene

- C) 3-propyl-3-hexene
D) 4-ethyl-4-hexene

Solution : (Correct Answer: A)

According to IUPAC:

Longest carbon chain consists of 6 carbons. Hence, it is hexene.

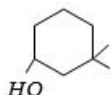
Closest double bond from starting is from second carbon. Hence it is hex-2-ene.

Now, ethyl group C_2H_5 is attached at the third carbon from starting.

Hence it is 3-ethyl-2-hexene.

Hence, IUPAC name is 3-ethyl-2-hexene.

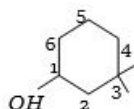
89. The IUPAC name of the compound is



- A) 1,1-dimethyl-3-hydroxy cyclohexane
B) 3,3-dimethyl-1-cyclohexanol
C) 3,3-dimethyl-1-hydroxy cyclohexane
D) 1,1-dimethyl-3-cyclohexanol

Solution : (Correct Answer: B)

(a) 3,3-dimethyl-1-cyclohexanol



90. IUPAC name of the compound is $CH_3 - \underset{\begin{array}{c} | \\ CH_2 \\ | \\ CH_3 \end{array}}{CH} - CH_2 - CH(OH) - CH_3$ is

- A) 4-ethyl-2-pentanol
B) 4-methyl-2-hexanol
C) 2-ethyl-2-pentanol
D) 3-methyl-2-hexanol

Solution : (Correct Answer: B)

Longest carbon chain has 6 atoms. We number from right hand side as alcohol group at C-2 gets least number in this case and suffix used for alcohol is "-ol". Methyl group is present at C-4 position.

Thus, IUPAC name is 4-Methyl-2-hexanol.

Biology - (Zoology)

91. Silencing of specific mRNA is possible via RNAi because of...

- A) Complementary dsRNA
B) Inhibitory ssRNA
C) Complementary tRNA
D) Non-complementary ssRNA

Solution : (Correct Answer: A)

RNAi (RNA interference) takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

92. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

- A) Bacterium
B) Yeast

- C) Virus
D) Phage

Solution : (Correct Answer: A)

The correct answer is bacterium.

In 1983, Eli Lilly, an American company, prepared two DNA sequences corresponding to 'A' and 'B' chains of human insulin and introduced them in plasmids of E.coli (a gram negative bacterium) to produce insulin chains.

93. Polymerase chain reaction (PCR) amplifies DNA following the equation.

- A) N^2
B) 2^n
C) $2n + 1$
D) $2 N^2$

Solution : (Correct Answer: B)

PCR i.e., polymerase chain reaction amplifies DNA as per the equation 2^n , where 'n' refers to number of cycles.

Thus, say, if 3 PCR cycles will run, then 2^3 i.e., $2 \times 2 \times 2 \Rightarrow 8$ DNA fragments will be formed.

94. Which of the following enzyme(s) are NOT essential for gene cloning?

- A. Restriction enzymes
B. DNA ligase
C. DNA mutase
D. DNA recombinase
E. DNA polymerase

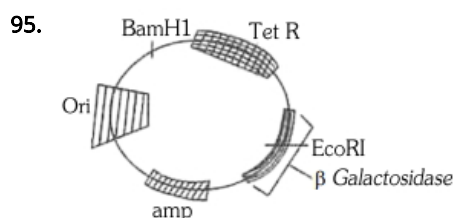
Choose the correct answer from the options given below:

- A) C and D only
B) A and B only
C) D and E only
D) B and C only

Solution : (Correct Answer: A)

Gene cloning is a process where a specific gene or DNA sequence is isolated and replicated, creating multiple identical copies.

In gene cloning, restriction enzymes, DNA ligase and DNA polymerase are primarily used.



In the above represented plasmid, an alien piece of DNA is inserted at EcoRI site. Which of the following strategies will be chosen to select the recombinant colonies?

- A) Using ampicillin & tetracycline containing medium plate.
B) Blue color colonies will be selected.
C) White color colonies will be selected.
D) Blue color colonies grown on ampicillin plates can be selected.

Solution : (Correct Answer: C)

The correct answer is that white-colored colonies will be selected.

Since an alien piece of DNA is being inserted at EcoRI site, the gene β -galactosidase present here will undergo insertional inactivation.

This gene is responsible for producing blue-colored colonies, but since it has been insertional inactivated, white colored colonies will be produced.

Ampicillin and tetracycline resistance genes present in the given DNA will remain intact. Thus, the given DNA will show amp^R and tet^R.

96. The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.

Given below are two statements about this method:

Statement I : The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Statement II : The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

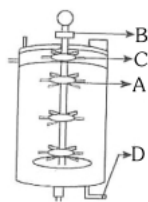
In the light of the above statements, choose the most appropriate answer from the options given below:

- A) Both Statement I and Statement II are correct
 B) Both Statement I and Statement II are incorrect
 C) Statement I is correct but Statement II is incorrect
 D) Statement I is incorrect but Statement II is correct

Solution : (Correct Answer: D)

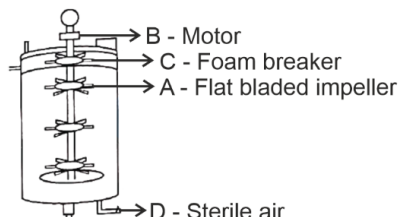
Statement I is incorrect but statement II is correct as a recombinant DNA is inserted within the coding sequence of an enzyme, β -galactosidase. This results into inactivation of the gene for synthesis of this enzyme. Thus, presence of insert results into insertional inactivation of the β -galactosidase gene and the colonies do not produce any colour and identified as recombinant colonies. Whereas non-recombinant transformants will produce blue colour in presence of chromogenic substrate.

97. Identify the part of a bio-reactor which is used as a foam braker from the given figure.



- A) A B) B C) D D) C

Solution : (Correct Answer: D)



∴ Part labelled as C is foam breaker.

98. Given below are two statements :
 Statement I: The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II : Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the most appropriate answer from the options given below :

- A) Both statement I and statement II are correct
 B) Both statement I and statement II are incorrect
 C) Statement I is correct but statement II is incorrect
 D) Statement I is incorrect but statement II is correct

Solution : (Correct Answer: A)

The cutting of DNA by restriction endonucleases results in the fragments of DNA. These fragments can be separated by a technique known as gel electrophoresis.

The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution. The DNA fragments purified in this way are used in constructing rDNA by joining them with cloning vectors.

- In gel electrophoresis, the DNA fragments separate (resolve) according to their size through sieving effect provided by the agarose gel. Hence, the smaller the fragment size, the farther it moves from cathode towards anode.

99. Which of the following is an example of non-distilled alcoholic beverage produced by yeast?

- A) Whisky B) Brandy C) Beer D) Rum

Solution : (Correct Answer: C)

Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of fermented broth.

100. Identify the statement that is NOT correct.

- A) Each antibody has two light and two heavy chains.
 B) The heavy and light chains are held together by disulfide bonds.
 C) Antigen binding site is located at C-terminal region of antibody molecules.
 D) Constant region of heavy and light chains are located at C-terminus of antibody molecules

Solution : (Correct Answer: C)

Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H₂ L₂.

In an antibody molecule, antigen binding site is located at N-terminal region.

101. Choose the true statements

I. Innate immunity is accomplished by providing different types of barriers

II. Acquired immunity is present from the birth and is inherited from parents

III. Sweat, tears, acid in the stomach and saliva prevent microbial growth

- A) I–True II–False III–True
 B) I–True II–True III–False
 C) I–False II–False III–True
 D) I–False II–True III–False

Solution : (Correct Answer: A)

Acquired immunity is not present from the birth and develops during the individual's lifetime. It is pathogen specific

102. Modern lifestyle is responsible for

- A) Lowering of immunity
 B) More sensitivity of allergens
 C) Both (a) and (b)
 D) None of the above

Solution : (Correct Answer: C)

Modern-day lifestyle has resulted in lowering of immunity and more sensitivity to allergens. More and more children in metro cities of India suffer from allergies and asthma due to sensitivity to the environment

103. Heroin is obtained from plant of family

- A) Papaveraceae B) Leguminosae
 C) Cruciferae D) Liliaceae

Solution : (Correct Answer: A)

Heroin is diamorphine or diacetylmorphine. It is a semi-synthetic opiate, derived from opium, which is a dried latex of unripe capsular fruits of poppy plant, *Papaver somniferum* of family-Papaveraceae

104. Which one of the following statements is correct?

- A) Patients, who had undergone surgery are given cannabinoids to relieve pain
 B) Benign tumors show the property of metastasis
 C) Heroin accelerates body functions
 D) Malignant tumors may exhibit metastasis

Solution : (Correct Answer: D)

Malignant tumor first grows slowly. No symptoms are noticed. This stage is called the latent stage. The tumour later grows quickly. The cancer cells go beyond adjacent tissue and enter the blood and lymph. Once this happens, they migrate to many other sites in the body, where the cancer cells continue to divide. It is metastasis. Only malignant tumors are properly designated as cancer.

105. Nicotine intake stimulates the ...A... to release ...B... and ...C... into blood circulation. This leads to increase in ...D... and an increase ...E... Here A and E refers to

- A) A–adrenal gland, B–adrenaline, C–nor-adrenaline, D–blood pressure, E– heart rate
 B) A–thyroid gland, B–thyroxine, C–parathyroxine, D–blood pressure, E– heart rate
 C) A–adrenal gland, B–thyroxine, C–nor-adrenaline, D–blood pressure, E–heart rate
 D) A–gonads, B–adrenaline, C–nor-adrenaline, D–blood pressure, E– heart rate

Solution : (Correct Answer: A)

A–Adrenal gland, B–Adrenaline, C–Nor-adrenaline, D–Blood pressure, E–Heart rate.

Tobacco contains a large number of chemical substances including nicotine, an alkaloid. Nicotine stimulates adrenal gland to release adrenaline and noradrenaline into blood circulation, both of which raise blood pressure and increase heart rate.

Smoking is associated with increased incidence of cancers of lung, urinary bladder and throat, bronchitis, emphysema, coronary heart diseases, gastric ulcer, etc.

106. An autoimmune disease is

- A) AIDS B) Haemophilia
 C) Allergy D) Myasthenia gravis

Solution : (Correct Answer: D)

Myasthenia gravis is an autoimmune disease characterized by the chronic fatigability and weakness of muscles, especially in the face and throat, as result of defect in the conduction of nerve impulses at the myoneural junction.

107. Match the following diseases with the causative organism and select the correct option.

Column –I	Column –II
(a) Typhoid	(i) Wuchereria
(b) Pneumonia	(ii) Plasmodium
(c) Filariasis	(iii) Salmonella
(d) Malaria	(iv) Haemophilus

(a) (b) (c) (d)

- A) (iv) (i) (ii) (iii)
 B) (i) (iii) (ii) (iv)
 C) (iii) (iv) (i) (ii)
 D) (ii) (i) (iii) (iv)

Solution : (Correct Answer: C)

(iii) (iv) (i) (ii)

108. Which of the following disease caused internal bleeding, muscular pain, fever, anaemia and blockage of the intestinal passage?

- A) Ascariasis B) Filariasis
 C) Amoebiasis D) Trypanosomiasis

Solution : (Correct Answer: A)

Symptoms of ascariasis disease includes internal bleeding, muscular pain, fever anaemia and blockage of the intestinal passage. Ascariasis is caused by an endoparasite of human, *Ascaris lumbricoides* commonly called roundworm

109. Antigen is a substance which

- A) lowers body temperature
 B) destroys harmful bacteria
 C) triggers the immune system
 D) is used as an antidote to poison

Solution : (Correct Answer: C)

Antigen is the only foreign substance which triggers immune system by way of antibody formation. These antibodies neutralize antigen by antigen-antibody reaction.

110. The tests that are used in the diagnosis of AIDS are

- A) ELISA and immunoblot
 B) Northern blot and ELISA
 C) ELISA and southern blot

D) Western blot and ELISA

Solution : (Correct Answer: D)

The two common tests used in the diagnosis of AIDS are Western Blot (culturing of blood and testing plasma for virus) and ELISA (Enzyme Linked Immunosorbent Assay).

111. A - The vaccines generate memory - B and T - cells. R - In vaccination activated pathogens are introduced into the body.

- A) A and R both are correct.
B) A and R both are incorrect.
C) A is correct, R is incorrect.
D) A is incorrect, R is correct.

Solution : (Correct Answer: C)

112. Assertion : Cocaine has a potent stimulating action on central nervous system, producing a sense of euphoria and increased energy.

Reason : Injecting the microbes intentionally during immunisation or infectious organisms gaining access into body during natural infection induces active immunity.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
C) If the Assertion is correct but Reason is incorrect.
D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: C)

Cocaine is obtained from the cocapant, *erthyroxylum coca*. Cocaine is commonly called as coke or crack which is usually snored. It has a potent stimulating action of euphoria and increased energy. It interferes with the transport of the neurotransmitter dopamine. Its increased dosages cause hallucinations.

113. Which is the particular type of drug that is obtained from the plant whose one flowering branch is shown here?



- A) Hallucinogen B) Depressant
C) Stimulant D) Pain killer

Solution : (Correct Answer: A)

(a) : The plant illustrated in the diagram is *Datura*. Seeds of *Datura stramonium* are misused for their hallucinogenic properties because of the presence of anticholinergic alkaloids atropine, hyoscyamine and scopolamine (= hyoscine). However, even in slight excess, they can cause death.

114. Which one of the following non-infectious disease is the major cause of death in humans ?

- A) Cancer B) AIDS C) Asthma D) Typhoid

Solution : (Correct Answer: A)

115. Anamnestic response

- A) Encounter a pathogen first time

B) Encounter pathogen second time

C) A and B both

D) None of the above

Solution : (Correct Answer: B)

116. Symptoms of amoebiasis

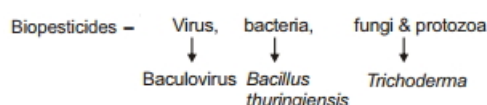
- A) Abdominal pain, cramps, stools with excess mucous and blood clots and constipation
B) Internal bleeding, muscular pain, fever, anemia
C) Inflammation of legs, mutation in genital organs
D) Fever, chills, Headache, cough

Solution : (Correct Answer: A)

117. Which of the following is included in biopesticide?

- A) Viruses and bacteria only
B) Viruses, bacteria and fungi only
C) Viruses, bacteria, fungi and protozoa
D) Viruses, bacteria, fungi and protozoa only

Solution : (Correct Answer: C)



118. __A__ enzyme is used to remove clots from the blood vessels of patients who have undergone __B__ leading to heart attack.

- A) A—Streptokinase; B—Myocardial infarction
B) A—Lipases; B—Arteriosclerosis
C) A—Proteases; B—Myocardial infarction
D) A—Pectinases; B—Atherosclerosis

Solution : (Correct Answer: A)

A—Streptokinase; B—Myocardial infarction

119. Organic farming includes

- I. use of biofertilisers and biopesticides
II. crop rotation
III. locally developed pest resistant varieties
Choose the correct option

- A) I and II
B) I and III
C) II and III
D) I, II and III

Solution : (Correct Answer: D)

Chemical fertilisers cause pollution of water bodies as well as ground water, besides getting stored in crop plants. Therefore, environmental scientist are pressing for switch over to organic farming. Organic farming is a form of agriculture that relies on techniques such as crop rotation, green manure, compost and biological pest control

120. Yeast have been used for the commercial production of

- I. ethanol II. bread III. cheese

Choose the correct option

- A) I and II B) I and III
C) I, II and III D) None of these

Solution : (Correct Answer: A)

Alcoholic beverages are defined as beverages that contain ethanol (C_2H_5OH). This ethanol is almost always produced by fermentation, the metabolism of carbohydrates by certain species of yeast under anaerobic or low-oxygen conditions. Beverages

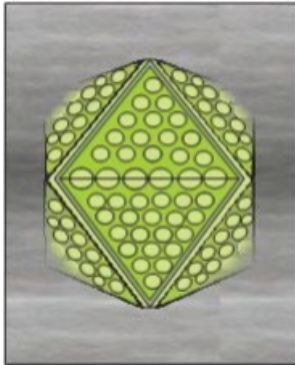
such as, wine, beer, or distilled spirits all use yeast at some stage of their production.

Yeast the most common one being *Sacharomyces cerevisiae*, is used in baking as leavening agent, where it converts the food/fermentable sugars present in dough into the gas carbon dioxide. This causes the dough to expand or rise as gas forming pockets or bubbles. When the dough is baked, the yeast dies and the air pockets 'set', giving the baked product a soft and spongy textures.

Cheese is formed by partial degradation of milk by different other microorganisms

121. Diagram of Adenovirus

Choose the correct for the given diagram



- A) Adenovirus - Cause of respiratory disease
- B) Bacteriophage - Tobacco mosaic disease
- C) Viroid - Potato spindle tumor disease
- D) Prion - Alzheimer's disease

Solution : (Correct Answer: A)

The diagram is of Adenovirus, which is the causative agent of respiratory diseases.

122. A patient brought to a hospital with myocardial infarction is normally immediately given

- A) penicillin
- B) streptokinase
- C) cyclosporin- A
- D) statins.

Solution : (Correct Answer: B)

(b) : Streptokinase (Tissue Plasminogen Activator or *TPA*) is an enzyme obtained from the culture of some haemolytic bacterium *Streptococcus* which is modified genetically to function as clot buster. Therefore, it helps in clearing blood clots inside the blood vessels through dissolution of intravascular fibrin during myocardial infarction.

123. Primary treatment is the

- A) physical removal of large and small particles from sewage
- B) biological removal of large and small particles from sewage
- C) Both (A) and (B)
- D) chemical removal of large and small particles from sewage

Solution : (Correct Answer: A)

124. Priyons are....

- A) Fungi with lipids
- B) Proteineous microbes
- C) Algae with carbohydrates
- D) None

Solution : (Correct Answer: B)

125. Insecticides kill

- A) Harmful insects
- B) Only plant pests
- C) Specific insects
- D) Both harmful and useful insects

Solution : (Correct Answer: D)

(d) Pesticides kill both harmful and useful insects.

126. *VAM* represents

- A) Saprophytic fungi
- B) Symbiotic fungi
- C) Saprophytic bacteria
- D) Symbiotic bacteria

Solution : (Correct Answer: B)

It's obvious.

127. '*ELISA*' test is done for the diagnosis of

- A) Anthrax
- B) *HIV*
- C) Hepatitis
- D) Malaria

Solution : (Correct Answer: B)

It's obvious.

128. Chemicals, which can induce cancer are called

- A) Mutagenic agents and produce benign tumour
- B) Carcinogens and produce non-malignant tumour
- C) Mutagenic agents and do not produce malignant tumour
- D) Carcinogens and produce malignant tumour

Solution : (Correct Answer: D)

It's obvious.

129. Cell division or mitosis is a normal process in a living cell, but sudden and abnormal mitosis in an organ will frequently result in a

- A) Zygote
- B) Gastrula
- C) New organ
- D) Cancer

Solution : (Correct Answer: D)

(d) The cancer cells multiply by abnormal rapid and uncontrolled mitosis.

130. Man in the life cycle of *Plasmodium* is

- A) Primary host
- B) Secondary host
- C) Intermediate host
- D) None of these

Solution : (Correct Answer: A)

It's Obvious

131. Sensitivity to any allergen is related to

- A) Deviation from the process of immunity
- B) Age of the person
- C) Eating habit
- D) Rise in environmental temperature

Solution : (Correct Answer: A)

It's obvious.

132. Infectious or communicable diseases

- A) Cannot be communicated from one man to another
- B) Can develop infection from one man to another
- C) Can spread without any intermediate host
- D) Can spread with some intermediate host

Solution : (Correct Answer: B)

It's obvious.

133. Which one of the following sets includes the bacterial disease

- A) Cholera, typhoid, mumps
- B) Tetanus, tuberculosis, measles
- C) Malaria, mumps, poliomyelitis
- D) Diphtheria, leprosy, plague

Solution : (Correct Answer: D)

It's obvious.

134. The biological agents of disease include

- A) Minerals, vitamins, proteins and carbohydrates
- B) Viruses, bacteria, fungi, helminths and other organisms
- C) Heat, cold, humidity pressure, radiations
- D) All the above

Solution : (Correct Answer: B)

It's obvious.

135. Malarial parasites could be best obtained from a patient

- A) An hour before rise of temperature
- B) When temperature rises with rigor
- C) When temperature comes to normal
- D) Few hours after the temperature reaches to normal

Solution : (Correct Answer: D)

It's obvious.

Biology - (Botany)

136. Basmati is unique for its aroma and flavour, whose *A*... varieties are cultivated in *B*.... Here *A* and *B* refers to

- A) *A* – 27; *B*—America
- B) *A* – 30; *B*—America
- C) *A* – 27; *B*—India
- D) *A* – 30; *B*—India

Solution : (Correct Answer: C)

Basmati is unique for its aroma and flavor, whose 27 varieties are cultivated in India

137. With regard to insulin choose correct options.

- (a) *C*–peptide is not present in mature insulin.
- (b) The insulin produced by *rDNA* technology has *C*–peptide.
- (c) The pro-insulin has *C*–peptide.
- (d) *A* –peptide and *B* –peptide of insulin are interconnected by disulphide bridges.

Choose the correct answer from the options given below.

- A) (b) and (d) only
- B) (b) and (c) only
- C) (a), (c) and (d) only
- D) (a) and (d) only

Solution : (Correct Answer: C)

138. Identify the incorrect statement.

- (i) *Bacillus thuringiensis* are available in sachets as dried spores which are mixed with kerosine and sprayed on to vulnerable plants
- (ii) *B. thuringiensis* are used to control butterfly caterpillars, but leave other insects unharmed
- (iii) The toxin of *B. thuringiensis* is released in the blood of larvae and the larvae get killed
- (iv) By the development of methods of genetic engineering *B. thuringiensis* toxin genes have been introduced

- A) (i) and (ii)
- B) (ii) and (iii)
- C) (i) and (iii)
- D) (ii) and (iv)

Solution : (Correct Answer: C)

(i) Dried spores are mixed with H_2O

(ii) Toxin of *Bacillus thuringiensis* is released in the gut of larvae and the larvae get killed.

139. Which of the following is correctly matched?

- A) *Agrobacterium tumefaciens* - Tumour
- B) *Thermos aquaticus* - Bt-gene
- C) *pBR322* – Enzyme
- D) Ligase - Molecular scissors

Solution : (Correct Answer: A)

Agrobacterium tumefaciens (updated scientific name: *Rizobium radiobacte*) is the causal agent of crown gall disease (the formation of tumour) in over 140 species of dicot. It is a rod-shaped, Gram negative soil bacterium (Smith, et. al, 1907).

Symptoms are caused by the insertion of a small segment of *DNA*, known as *T – DNA* (transfer *DNA*) into the plant cell, which is incorporated at a semi-random location into the plant genome.

140. Which one of the following molecular diagnostic technique is used to detect the presence of a pathogen in its early stage of infection?

- A) Angiography
- B) Radiography
- C) Enzyme replacement technique
- D) Polymerase Chain Reaction (*PCR*)

Solution : (Correct Answer: D)

Presence of a pathogen (bacteria, viruses, etc.) is usually suspected only when the pathogen has produced a diseased symptom. By that time the number of pathogens is already very high in the body but very low count of a bacteria or virus (when the symptoms of the disease are not yet visible) can be detected by multiplication of their nucleic acid by *PCR*

141. Silencing of a gene could be achieved by the use of

- A) *RNAi*
- B) Antisense *RNA*
- C) Both (a) and (b)
- D) None of these

Solution : (Correct Answer: C)

Silencing of a gene could be achieved through the use of *RNAi* and antisense *RNA*

142. Which of the following shows correct chronological order of the events occurring during callus culture?

- A) Callus → Cell division → Explant → Addition of cytokinin → Cells acquire meristematic property
- B) Explant → Callus → Cell division → Addition of cytokinin → Cells acquire meristematic property
- C) Explants → Cell division → Callus → Addition of cytokinin → Cells acquire meristematic property
- D) Callus → Explant → Cell division → Addition of cytokinin → Cells acquire meristematic property

Solution : (Correct Answer: C)

In callus culture, cell division in explant forms a callus. Callus is irregular unorganized and undifferentiated mass of actively dividing cells. Darkness and solid medium gelled by agar stimulates callus formation. The culture medium contains growth regulators auxin 2, 4 – D and often a cytokinin like BAP. Both of these growth regulators stimulate meristematic property in callus.

143. A : Transgenic mice are being used to test the safety of the polio vaccine.

R : It could replace the use of monkeys to test the safety of batches of the vaccine.

- A) Assertion and Reason both are correct and also correct explanation.
- B) Assertion and Reason both are correct but not explanation of assertion.
- C) Assertion is correct, but Reason is incorrect.
- D) Both Assertion and Reason are incorrect.

Solution : (Correct Answer: B)

144. A : ELISA can enable very easy detection of infection through antigen antibody interaction.

R : It is a nucleic acid based diagnostic tool that can confirm presence of infectious microbe at early stages.

- A) Assertion and Reason both are correct and also correct explanation.
- B) Assertion and Reason both are correct but not explanation of assertion.
- C) Assertion is correct, but Reason is incorrect.
- D) Both Assertion and Reason are incorrect.

Solution : (Correct Answer: C)

It is an antigen-antibody reaction based diagnostic tool that can confirm presence of infectious microbe at early stage.

145. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection?

- A) Western Blotting Technique
- B) Southern Blotting Technique
- C) ELISA Technique
- D) Hybridization Technique

Solution : (Correct Answer: C)

146. Which of the following are fused in somatic hybridization involving two varieties of plants?

- A) Somatic embryos B) Protoplasts
- C) Pollens D) Callus

Solution : (Correct Answer: B)

Protoplast of two varieties of plants are fused in somatic hybridization.

147. Consider the following statements about insulin.

- I. Human insulin is made-up of 51 amino acids arranged in two polypeptide chains.
- II. The two polypeptide chains are interconnected by two disulphide bridges.
- III. In mammals including humans, insulin is synthesised as a pro-hormone, which contains an extra stretch called the C-peptide.
- IV. C-peptide is not present in the mature insulin

Which of the statements given above are correct?

- A) I, II and III only
- B) I, II and IV only
- C) I, III and IV only
- D) I, II, III and IV

Solution : (Correct Answer: D)

148. What triggers activation of protoxin to active Bt toxin of Bacillus thuringiensis in boll worm?

- A) Body temperature
- B) Moist surface of midgut
- C) Alkaline pH of gut
- D) Acidic pH of stomach

Solution : (Correct Answer: C)

149. Silencing of mRNA has been used in producing transgenic plants resistant to

- A) bollworms B) nematodes
- C) white rusts D) bacterial blights.

Solution : (Correct Answer: B)

(b)

150. Assertion : Insect resistant transgenic cotton has been produced by inserting Bt gene.

Reason : The Bt gene is derived from a bacterium.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: B)

Bt— cotton is a transgenic crop. Transgenic plants are those plants, which have foreign gene incorporated in their DNA. This insect resistant gene is derived from a bacterium, Bacillus thuringiensis.

151. GMO technology is useful for

- A) Making crops more tolerant to abiotic stresses
- B) Helping to reduce post-harvest losses
- C) Enhancing nutritional value of food
- D) All the above

Solution : (Correct Answer: D)

152. Which body of the Government of India regulates GM research and safety of introducing GM organisms for public services ?

- A) Genetic Engineering Approval Committee

- B) Research Committee on Genetic Manipulation
- C) Bio-safety committee
- D) Indian Council of Agricultural Research

Solution : (Correct Answer: A)

(a) : Genetic modification of organisms can have unpredictable results, when such organisms are introduced into the ecosystem. Therefore, the Indian Government has set up organizations such as *GEAC* (Genetic Engineering Approval Committee), which makes decisions regarding the validity of *GM* research and the safety of introducing *GM* organisms for public services.

153. Genetic engineering is the

- A) Formation of new gene artificially
- B) Formation of *RNA* from *DNA* artificially
- C) Modification of genes artificially
- D) Formation of *DNA* from non *DNA* material

Solution : (Correct Answer: C)

(c) Genetic engineering is *DNA* manipulation technology that produces and modified *DNA* sequences for creating cells, tissues organs and individuals to suit human requirements.

154. A..... infects the roots of tobacco plants which reduce the production of tobacco.

- A) Nematode (*Meloidogyne incognita*)
- B) Coleopterans (beetles)
- C) Lepidopterns (armyworm)
- D) Dipterans (Mosquitoes)

Solution : (Correct Answer: A)

155. The crops engineered for glyphosate are resistant/tolerant to

- A) insects
- B) herbicides
- C) fungi
- D) bacteria.

Solution : (Correct Answer: B)

(b) : Glyphosate is a broad spectrum herbicide which especially kills broad leaved herbs. Crop plants may also get affected by the herbicide, thus now crop plants are genetically engineered for glyphosate resistance. So, when glyphosate herbicide is applied, only weeds and no crop plants get harmed.

156. It is now possible to breed plants and animals with desired characters through

- A) Genetic engineering
- B) Chromosome engineering
- C) Ikebana technique
- D) Tissue culture

Solution : (Correct Answer: A)

It's Obvious

157. *Bacillus thuringiensis* is a good

- A) Biofertilizer
- B) Biopesticide
- C) Biofuel
- D) Single cell protein

Solution : (Correct Answer: B)

(b) *Bacillus thuringiensis* produces a Biopesticide called thurioside. The bacterium is also sprayed as spores to control insect pests.

158. The "Ti plasmid" of *Agrobacterium tumefaciens* stands for

- A) Tumor independent plasmid
- B) Tumor inducing plasmid
- C) Temperature independent plasmid

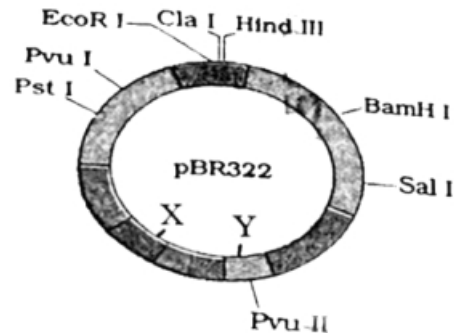
- D) Tumour inhibiting plasmid

Solution : (Correct Answer: B)

The correct answer is option (2) as Ti plasmid of *Agrobacterium tumefaciens* is tumor inducing plasmid, containing *T-DNA* which causes tumor in several dicot plants.

Options (1), (3) and (4) are not correct.

159. The following diagram showing restriction sites in *E. coli* cloning vector pBR322. Find the role of 'X' and 'Y' genes:



- A) The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.
- B) The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.
- C) Gene 'X' is responsible for recognitions sites and 'Y' is responsible for antibiotic resistance
- D) The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.

Solution : (Correct Answer: A)

Correct answer is option (1), because

'X' in the given diagram is ori while 'Y' is rop.

'X' which is ori is responsible for controlling the copy number of the linked DNA and 'Y' which is rop codes for protein involved in the replication of plasmid.

Options (2), (3) and (4) are incorrect as 'X' and 'Y' are not related to these functions.

160. Natural genetic engineer is:

- A) *Bacillus subtilis*
- B) *Pseudomonas* spp
- C) *Escherichia coli*
- D) *Agrobacterium tumefaciens*

Solution : (Correct Answer: D)

161. Cutting of a piece of *DNA* from a plasmid was done with the help of ...A... enzymes, popularly known as ...B.... Here A and B can be

- A) A—Tu ligases; B— Molecular glu
- B) A—Restriction enzyme; B— Molecular scissors
- C) A—Joining enzyme; B— Molecular glu
- D) A— *DNA* polymerases; B—Synthesising enzymes

Solution : (Correct Answer: B)

Cutting of piece of *DNA* from a plasmid was done with the help of restriction enzyme, popularly known as molecular scissors

162. In bacteria, genes for antibiotic resistance are usually located in:

- A) Chromosomal *DNA* B) Cytoplasm
C) Mitochondria D) Plasmids

Solution : (Correct Answer: D)

163. Which cannot be used for direct gene transfer

- A) Biolistics (gene gun)
B) Microinjection
C) Electroporation
D) *Agrobacterium tumefaciens*

Solution : (Correct Answer: D)

Biolistic, microinjection, and electroporation are the methods of direct gene transfer.

164. Choose the correct pair from the following

- A) Exonucleases : Make cuts at specific positions within *DNA*
B) Ligases : Join the two *DNA* molecules
C) Polymerases : Break the *DNA* into fragments
D) Nucleases : Separate the two strands of *DNA*

Solution : (Correct Answer: B)

Ligases : Join the two *DNA* molecules

165. Due to chloramphenicol resistance gene, one is able to select a transformed cell in the presence of chloramphenicol. The chloramphenicol resistance gene in this case is called

- A) Origin of replication
B) Selectable marker
C) Cloning sites
D) Insertional inactivation

Solution : (Correct Answer: B)

Due to chlorophenicol resistance gene, one is able to select a transformed cell in the presence of chloramphenicol. The chloramphenicol resistance gene in this case is called selectable marker

166. *DNA* can be introduced into any cell by:

- A) Injection
B) Being complexed with calcium salts
C) Being placed along with the cell into a gene gun
D) Gel electrophoresis

Solution : (Correct Answer: B)

167. In recombinant *DNA* technique, the term vector refers to a

- A) Donor *DNA*, it is identified and picked up through electrophoresis
B) Plasmid, transfers *DNA* into living cell
C) Collection of entire genome in form of plasmid
D) Enzyme, cuts the *DNA* at specific sites

Solution : (Correct Answer: B)

Vector is a plasmid or virus *DNA* used to introduce genes into a host cell, where the genes may be amplified (gene cloning) or otherwise manipulated

168. Which of the following steps is/are catalysed by Taq polymerase in a *PCR*?

- A) Denaturation of template *DNA*
B) Annealing of primers to template *DNA*
C) Extension of primer end on template *DNA*
D) All of these

Solution : (Correct Answer: C)

Extension of primer is done by adding nucleotides to the primer which is done by *DNA* polymerase

here taq polymerase.

169. *RNA* is removed by the treatment with

- A) Ribonuclease B) Protease
C) Chitinase D) Cellulase

Solution : (Correct Answer: A)

RNA is removed by treatment with ribonuclease

170. The components of a bioreactor are

- I. an agitator system
II. an oxygen delivery system
III. foam control system
IV. temperature control system
V. *pH* control system
VI. sampling ports to withdraw cultures periodically
Choose the correct option

- A) I, II, III, IV and V
B) II, IV, V and VI
C) I, II, III, IV and VI
D) All of these

Solution : (Correct Answer: D)

Components of a bioreactors. An agitator system. An oxygen delivery system. Foam control system. Temperature control system. *pH* control system. sampling ports to withdraw culture periodically

171. Gene for cloning may be chemically synthesized:

- A) When the exact sequence of nucleotides is known
B) Through the use of restriction enzymes and gel electrophoresis to separate restriction fragments
C) By the Sanger method
D) By making complementary *DNA* from genes without introns

Solution : (Correct Answer: A)

172. Which of the following statement is not correct regarding *Eco RI* restriction endonuclease enzyme?

- I. *Eco. RI* restriction endonuclease enzyme is isolated from *Escherichia coli* *RY13*
II. Its recognition sequence is 5' – *GAATTC* – 3'
3' – *CTTAAG* – 5'
5' – *G – A – A – T – T – C* – 3'
3' – *C – T – T – A – A – G* – 5'
III. Its site of cleavage is

- A) I and II
B) I and III
C) I, II and III
D) None of the above

Solution : (Correct Answer: D)

All statements are correct

Restriction Enzymes	Source	Recognition Sequence and Site of Cleavage	Product
<i>Eco RI</i>	<i>Escherichia coli</i> <i>RY 13</i>	$ \begin{array}{c} \downarrow \\ 5'-G-A-A-T-T-C-3' \\ 3'-C-T-T-A-A-G-5' \\ \uparrow \end{array} $	$ \begin{array}{c} G \quad A-A- \\ \\ C-T-T-A- \\ \text{Sticky end} \end{array} $

173. Which of the following is the most accepted definition of biotechnology by European Federation of Biotechnology?

- A) Maintenance of sterile ambience for enabling growth of desired microbe/eukaryotic cell in large quantities
- B) Technique of using live organism or enzyme from organisms to produce products and processes useful to animals
- C) Process which use genetically engineered animals only on a large scale for benefit of mankind
- D) The integration of natural science and organisms cells, parts thereof and molecular analogues for products and services

Solution : (Correct Answer: D)

Fact.

174. Which one of the following statement is not true regarding gel electrophoresis technique?

- A) The separated DNA fragments are stained by using ethidium bromide.
- B) The presence of chromogenic substrate gives blue coloured DNA bands on the gel.
- C) Bright orange coloured bands of DNA can be observed in the gel when exposed to *UV* light.
- D) The process of extraction of separated DNA strands from gel is called elution.

Solution : (Correct Answer: B)

175. Most suitable method of introducing alien *DNA* into a plant cell is

- A) Biolistics
- B) Micro-injection
- C) Lipofection
- D) Heat shock method

Solution : (Correct Answer: A)

176. The specific palindromic sequence which is recognized by *EcoRI* is :

- A) 5' – GGATCC – 3'
3' – CCTAGG – 5'
- B) 5' – GAATTC – 3'
3' – CTTAAG – 5'
- C) 5' – GGAACC – 3'
3' – CCTTGG – 5'
- D) 5' – CTTAAG – 3'
3' – GAATTC – 5'

Solution : (Correct Answer: B)

5' – GAATTC – 3'

3' – CTTAAG – 5'

177. I. *DNA* being a hydrophilic molecule cannot pass through cell membranes.

II. The bacteria should be made competent to accept the *DNA* molecule.

The Correct option regarding the above statements is

- A) I is true, but II is false
- B) II is true, but I is false
- C) I and II are true
- D) I and II are false

Solution : (Correct Answer: C)

178. *Taq* polymerase is isolated from a

- A) Virus
- B) Bacteria
- C) Fungus
- D) Protozoans

Solution : (Correct Answer: B)

179. Recombinant *DNA* technology was discovered by

- A) Khorana
- B) Bateson and de Vries
- C) Watson
- D) Cohen and Boyer

Solution : (Correct Answer: D)

180. Restriction endonucleases are

- A) Synthesized by bacteria
- B) Present in mammalian cells for degradation of *DNA*
- C) Used in genetic engineering
- D) Used for in vitro *DNA* synthesis

Solution : (Correct Answer: C)

(c) Restriction endonuclease cuts *DNA* at very specific site and is used in genetic engineering.